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Surgical Logistics

From operating room organisation to patient-reported outcomes in breast cancer surgery

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Abstract

Background

Efficient use of finite healthcare resources is a central challenge in surgical care. At the same time, perioperative productivity is shaped by organisational conditions that may influence not only efficiency and costs, but also staff work environment and patient-reported outcomes. The overall aim of this thesis was to explore how organisational factors in surgical care relate to productivity, resource use, staff experience, and patient-reported outcomes.

Methods and Results

Study I was a retrospective observational study comparing total case time between two differently organised surgical departments (general surgery and cardiothoracic surgery) at a Swedish university hospital, where the same group of surgeons performed elective surgery for breast cancer and malignant melanoma. After adjustment for patient-, procedure-, and surgeon-related factors, total case time was on average 30.7 minutes shorter at the cardiothoracic department ($p<0.001$).

Study II was a health-economic evaluation using a bottom-up micro-costing and cost-minimisation approach to compare breast cancer surgery at the same two departments. The cardiothoracic model was associated with lower mean cost per operation, greater procedural efficiency, and shorter simulated waiting times. At the observed annual volume, the point estimate corresponded to an annual cost saving of approximately SEK 681,000, although uncertainty remained around the per-operation cost difference.

Study III used a sequential explanatory mixed-methods design to examine how operating room nurses and anaesthesia nurses at the same two departments perceived productivity and what organisational conditions supported or thwarted a sustainable pace over time. Survey results showed significant between-unit differences in safety climate, stress recognition, and perceptions of management. The interviews identified four cross-cutting categories linked to the three basic psychological needs described in Self-Determination Theory: autonomy, competence, and relatedness. These were predictable workflow and usable tools, psychologically safe teamwork, participatory planning and

visible follow-through, and learning structures for mastery, with workload as a boundary condition.

Study IV was a longitudinal cohort study of 2,906 women using prospective patient-reported outcome measure (PROM) data from the Swedish National Quality Register for Breast Cancer to examine whether patients' diagnosis-to-surgery lead time was associated with worry about future health. Longer lead time was associated with greater worry about future health (adjusted odds ratio 1.15 per one standard deviation increase; p=0.008), but not with general worry, tension, depressed mood, or overall quality of life.

Conclusions

Taken together, the studies indicate that surgical organisation is associated with perioperative efficiency and costs as well as staff work environment and patient-reported outcomes. Organisational differences were reflected in total case time, resource use, perceived conditions for sustainable work, and patients' worry during the waiting period before surgery. These findings suggest that surgical productivity should not be understood solely as output or speed, but as a broader organisational phenomenon shaped by structures and processes that are relevant to both staff and patients. Improving surgical care therefore requires attention not only to efficiency, but also to sustainability, coordination, and timely access to treatment.

Keywords

operating room; surgical organisation; productivity; micro-costing; mixed-methods; Self-Determination Theory; breast cancer; patient-reported outcomes

Populärvetenskaplig sammanfattning på svenska

Bakgrund

Effektiv användning av hälso- och sjukvårdens begränsade resurser är en central utmaning inom kirurgisk vård. Samtidigt formas den perioperativa produktiviteten av organisatoriska förutsättningar som kan påverka inte bara effektivitet och kostnader, utan även personalens arbetsmiljö och patientrapporterade utfall. Det övergripande syftet med avhandlingen var att undersöka hur organisatoriska faktorer inom kirurgisk vård hänger samman med produktivitet, resursanvändning, personalens erfarenheter och patientrapporterade utfall.

Metod och resultat

Studie I var en retrospektiv observationsstudie där den totala perioperativa tiden jämfördes mellan två operationsavdelningar med olika organisatorisk utformning, en allmänkirurgisk och en thoraxkirurgisk, vid ett svenskt universitetssjukhus. Samma grupp kirurger utförde elektiv kirurgi för bröstcancer och malignt melanom vid båda avdelningarna. Efter justering för patient-, ingrepps- och kirurgrelaterade faktorer var den totala perioperativa tiden i genomsnitt 30,7 minuter kortare vid den thoraxkirurgiska avdelningen ($p<0,001$).

Studie II var en hälsoekonomisk utvärdering av bröstcancerkirurgi vid samma två avdelningar, baserad på mikrokostnadsanalys och en kostnadsminimeringsansats. Den thoraxkirurgiska modellen var förknippad med lägre genomsnittlig kostnad per operation, högre processeffektivitet och kortare simulerade väntetider. Vid den observerade årsvolymen motsvarade skattningen en årlig kostnadsbesparing på cirka 681 000 kronor, även om det fanns osäkerhet kring kostnadsskillnaden per operation.

Studie III använde en sekventiell förklarande design som kombinerade kvantitativa och kvalitativa metoder för att undersöka hur operationssjuksköterskor och anestesisjuksköterskor vid de två avdelningarna uppfattade produktivitet och vilka organisatoriska förutsättningar som stödde eller försvårade ett hållbart arbetstempo över tid. Enkäten visade signifikanta skillnader mellan enheterna avseende säkerhetsklimat, stressigenkänning och uppfattning om ledning. Intervjuerna identifierade fyra övergripande kategorier kopplade till autonomi, kompetens och samhörighet: förutsägbart arbetsflöde och

användbara verktyg, psykologiskt tryggt teamarbete, delaktighet i planering och synlig återkoppling samt lärandestrukturer för att utveckla kompetens, med arbetsbelastning som ett begränsande villkor.

Studie IV var en longitudinell kohortstudie med prospektivt insamlade patientrapporterade utfallsmått (PROM) från Nationellt kvalitetsregister för bröstcancer. Studien omfattade 2 906 kvinnor med primär invasiv bröstcancer och undersökte om ledtiden från diagnos till operation var associerad med oro för den framtida hälsan. Längre ledtid var associerad med större oro för den framtida hälsan (justerad oddskvot 1,15 för varje ökning av ledtiden med en standardavvikelse; p=0,008), men inte med generell oro, spänning, nedstämdhet eller övergripande livskvalitet.

Slutsats

Sammantaget visar studierna att den kirurgiska organisationen är förknippad inte bara med perioperativ effektivitet och kostnader, utan även med personalens arbetsmiljö och patientrapporterade resultat. Organisatoriska skillnader återspeglades i den totala perioperativa tiden, resursanvändningen, personalens upplevda förutsättningar för ett hållbart arbete och patienternas oro under väntan på operation. Kirurgisk produktivitet bör därför inte förstås enbart som produktion eller hastighet, utan som ett bredare organisatoriskt fenomen som formas av strukturer och processer med betydelse för både personal och patienter. Förbättring av kirurgisk vård kräver därmed uppmärksamhet inte bara på effektivitet, utan även på hållbarhet, samordning och tillgång till behandling i rätt tid.

List of papers

This thesis is based on the following published papers and manuscripts:

I. Eriksson J, Fowler P, Appelblad M, Lindholm L & Sund M. Productivity in relation to organisation of a surgical department: a retrospective observational study. BMC Surg. 2022 Mar 24;22(1):114. doi: 10.1186/s12893-022-01563-6

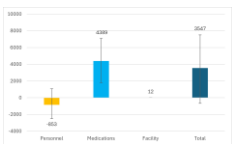
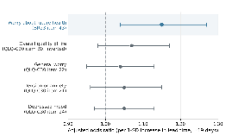
II. Eriksson J, Meili KW, Lindholm L, Appelblad M & Sund M. Health Economic Analysis of Organisational Models for Breast Cancer Surgery: A Bottom-Up Micro-costing and Cost-Minimisation Approach. Health Econ Rev. 2026 Feb 12;16(1):24. doi: 10.1186/s13561-026-00743-x

III. Eriksson J, Lämås K, Lindholm L, Appelblad M & Sund M. What Supports Sustainable Pace in the Operating Room? A Sequential Explanatory Mixed-Methods Study Using Self-Determination Theory. (*Conditionally Accepted*)

IV. Eriksson J, Fredriksson I, Lindholm L, Appelblad M & Sund M. Diagnosis-to-surgery lead time and future-health worry in women with primary breast cancer: a longitudinal Swedish registry-based cohort study. (*Under Review*)

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Thesis at a glance

	Aim	Method	Results	Conclusions
Study I 	Compare total case time between organisational models.	Retrospective observational study.	One organisational model reduced adjusted total case time by 31 min.	Organisation may improve perioperative efficiency.
Study II 	Compare costs, capacity, and waiting time implications.	Micro-costing and capacity modelling.	Lower costs, higher capacity, and shorter simulated waits.	Organisational configuration may improve cost-efficiency and capacity.
Study III 	Explore staff experiences of productivity and workflow.	Sequential explanatory mixed methods: SAQ-OR survey and qualitative content analysis of interviews.	Sustainable pace was linked to predictability, teamwork, participation, competence, and support.	Sustainable pace is organisationally conditioned.
Study IV 	Examine association between waiting time and patient-reported outcomes.	National registry-based PROM linkage.	Longer waiting time was associated with greater worry about future health.	Waiting time may contribute to cancer-related worry before surgery.

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Abbreviations

ALND	Axillary lymph node dissection
ASA	American Society of Anesthesiologists physical status classification
BCS	Breast conserving surgery
CHEERS	Consolidated Health Economic Evaluation Reporting Standards
CT	Cardiothoracic
EORTC	European Organisation for Research and Treatment of Cancer
FCOS	First case on-time start
GS	General surgery
ICU	Intensive care unit
KVÅ	Swedish procedure codes from the Classification of Health Care Measures, Swedish: Klassifikation av vårdåtgärder
NKBC	National quality register for breast cancer, Swedish: Nationellt kvalitetsregister för Bröstcancer
NOT	Non-operative time
OR	Operating room; odds ratio (Study IV)
PROMs	Patient-reported outcome measures
QLQ-BR23	Quality of Life Questionnaire Breast Cancer 23
QLQ-C30	Quality of Life Questionnaire Core 30
SAQ-OR	Safety Attitudes Questionnaire Operating Room
SLNB	Sentinel lymph node biopsy

SURG-TLX	Surgery task load index
SVF	Standardised care pathway, Swedish: Standardiserat Vårdförlopp
TCT	Total case time

Introduction

Surgical production and the role of the operating department

The operating theatre has historically been both a clinical and organisational space. In nineteenth-century surgery, the operating theatre was not only a site of treatment but also a space of observation, teaching, and professional performance. Toward the end of the nineteenth century, this public theatre model increasingly gave way to smaller and more controlled operating rooms (OR), partly driven by the need to separate septic from clean cases, restrict audiences, and reduce the risk of contamination (1). This historical transition is illustrated by the nineteenth-century operating theatre shown in Figure 1.



Figure 1. The Old Operating Theatre, London. Photograph by Mike Peel, licensed under CC BY-SA 4.0 via Wikimedia Commons.

In contemporary hospitals, this organisational dependence has intensified: modern operating departments rely on the synchronised functioning of surgical, anaesthetic, nursing, logistical, and postoperative resources. The operating department is therefore both a site of clinical

intervention and a key point at which hospital capacity, coordination, and productivity become visible.

Surgery is a central part of modern medicine and constitutes a large portion of hospitals' budgets. It has been estimated that about 234 million major surgical procedures are performed around the world each year (2). In Sweden, the number of surgical procedures in 2021 was about 405,000 in inpatients and 1.2 million in outpatients, and the trend is growing with 3 and 6.5 %, respectively, from the year before (3). A nationwide Swedish register study has also provided population-based estimates of both inpatient and outpatient surgical activity, illustrating the substantial volume and breadth of surgical care delivered across the healthcare system (4). The organisation of operating departments affects perioperative productivity, which is crucial for patient safety and hospitals' financial health and for society in general.

The OR is one of the most financially significant components of hospital operations. Some estimates suggest that it may generate as much as 40% of a hospital's revenue while accounting for nearly 30% of its expenditures. Accordingly, the operating department is regarded as one of the most economically important components of any hospital (5). Therefore, improving efficiency and time management in the OR is of major importance and involves not only streamlining processes but also avoiding any delays in starting surgeries.

Research has shown that operating rooms can become bottlenecks of the patient flow during high demand conditions, and that the management of surgical capacity directly impacts multiple downstream services. Importantly, decisions in the OR may have upstream and downstream effects. The capacity of downstream resources such as intensive care units (ICUs) and recovery rooms can actually constrain OR planning, creating a complex interdependency across the hospital system (6). One important factor in the OR that can bear such implications is the first case on-time start (FCOS). This metric has long been regarded as an indicator of efficiency for the OR schedule, as it helps to ensure that the daily schedule runs smoothly without any unnecessary disruptions. In order to optimise efficiency in the operating room, it is important to consider the potential cost savings and benefits that come with reducing idle time and improving patient and provider satisfaction. Cost-benefit analyses have shown that such measures can result in significant savings (7). Furthermore, strategies to address delays in surgery start times include implementing information technology to improve operational efficiency in the OR (8). Other small disturbances in the OR can have substantial cascading

consequences on patient flow and outcomes. Research demonstrates that even modest delays in surgical start times significantly impact patient safety and outcomes. Among 8,844 elective operations, when surgeons experienced waiting times exceeding one hour between consecutive operations, events that occurred in 71.4% of operations, there was a markedly increased frequency of major adverse events (13.9% versus 5.3% in operations without such delays). The risk of major adverse events increased by 37% when surgeons experienced scheduling delays (9). The mechanisms behind these cascade effects are multifaceted. Higher daily surgeon workload has been linked to both prolonged operative duration and longer postoperative intensive care stays (10). Even something as specific as patient transportation delays to the OR can trigger a chain reaction, where delays in transporting patients led to second-case cancellations and operating theatres running beyond scheduled hours, demonstrating how localised inefficiencies propagate through the system (11). Delayed access to operating rooms for emergency surgery has been associated with increased in-hospital mortality, longer hospital stays, and higher costs (12), underscoring that the bottleneck effect extends beyond operational metrics to directly impact patient outcomes and resource utilisation.

Beyond logistics, the human dimensions of perioperative work, including work climate, team communication, and staff motivation, play an important role in how surgical teams respond to and manage operational disruptions, suggesting that organisational conditions shape efficiency metrics as well as the resilience of the system as a whole.

Perioperative care as a complex organisational system

The pathway of the elective surgical patient is structured as a complex continuum, where each separate component needs to act in unison in order to facilitate an uninterrupted journey for the patient. It requires seamless interdisciplinary coordination and cooperation to ensure the best possible outcome of care (13). Perioperative care exemplifies this complexity particularly well, encompassing parts ranging from anaesthesia, nursing, administration, and surgical specialty to post-anaesthesia and recovery care (14) and is usually performed in a high-tech and complex environment where communication, collaboration and teamwork are key to successful outcomes (15).

What makes perioperative care organisationally distinct is the degree of interdependence between its constituent parts. Surgical production

depends not only on what happens in the operating room itself, but on the coordinated functioning of upstream and downstream processes, preoperative assessment and scheduling, anaesthesia preparation, availability of instruments and consumables, intensive care and recovery capacity, and postoperative ward resources. A constraint in any one of these components can propagate through the system, disrupting flow and reducing effective capacity (6). The interdependence between upstream, intraoperative, and downstream components of the surgical pathway is summarised in Figure 2.

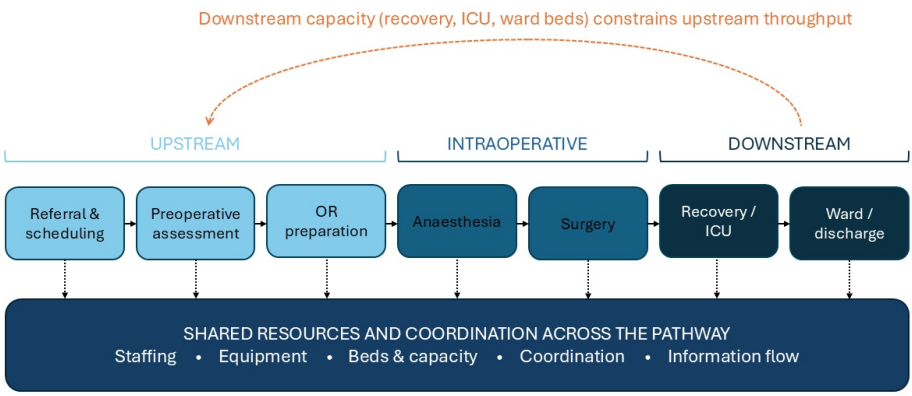


Figure 2. Simplified overview of the elective surgical pathway and the interdependence between upstream, intraoperative, and downstream resources.

Equally important is the recognition that variation and disruption are not exceptions to be eliminated, but inherent features of complex organisational systems. Staffing fluctuates, cases run longer than planned, equipment is unavailable, and patients present with unanticipated complexity. How a system is organised, its staffing structures, scheduling practices, communication norms, and resource allocation mechanisms, determines its capacity to absorb and respond to this variation without compromising safety or efficiency. In this sense, organisational form is not merely an administrative concern, but a direct determinant of how perioperative care functions in practice (9, 16).

Perioperative organisational systems must navigate several structural tensions simultaneously. Insufficient funding and staffing constrain a team's capacity to maintain both safety and efficiency, while pressure for increased productivity can, if poorly managed, erode the conditions necessary for high-quality care. Adequate training and present leadership are prerequisites rather than optional enhancements; without them, even

well-resourced teams struggle to perform consistently. Bureaucratic demands, including clinical adherence protocols and safety initiatives, compete with clinical time in ways that reduce uptake and effectiveness, and may benefit from being adapted to the specific context of individual surgical teams (15, 17-20).

These organisational factors interact, and their combined effects may either support or undermine safe and efficient perioperative care.

Productivity, efficiency and quality in surgical care

Productivity in surgical care is primarily measured through volume-based metrics. It refers to the number of surgeries performed per operating room each day, i.e. to the output of surgical services delivered (21). While useful, such measures provide only a partial account of surgical performance. To better capture dimensions such as case complexity, procedure duration, and unused operating room capacity, it is important to move beyond productivity alone and consider the broader concept of efficiency.

Efficiency, in this context, refers to process metrics and resource utilisation in surgical care delivery. It encompasses both OR output and the utilisation of available time. Overall, efficiency adds a temporal dimension to productivity. Important factors in predicting and evaluating operating room efficiency include total case time, reflecting the full duration of an individual surgical case, surgical incision start time, turnover time between cases, and in-hospital waiting time for surgery. Cancellation rates for elective surgical cases should also be considered. From a broader systems perspective, efficiency may also be understood in relation to patient need and the appropriate allocation of surgical resources. More broadly, it may be defined as the delivery of care using the least necessary resources while maintaining quality and safety. This also includes optimising workflow, reducing waste, and minimising delays in patient care delivery (21).

A related concept is throughput, which refers to the rate at which patients move through the surgical pathway. Strategies that optimise the proportion of OR time used for surgery have been shown to improve both efficiency and throughput, whereas constraints such as bed availability or postoperative care capacity may limit the number of cases completed regardless of intraoperative efficiency (22). High utilisation rates do not necessarily translate into high throughput, since reducing apparent idle

time too aggressively may remove the operational buffer needed to absorb variation in case duration and unplanned demand (23).

Cancellations constitute a particularly consequential efficiency failure, as they result in immediate loss of productive OR time. Cancellation rates in elective surgery vary widely but have been reported in the range of 10–35% in some settings, with OR downtime estimated to cost USD 30–50 per minute (24). The financial impact of each cancelled case has been estimated at USD 1,430–1,700, and cancellations also contribute to cascading schedule delays, reduced staff morale, and increased patient anxiety (25). Importantly, studies suggest that more than half of all cancellations are preventable, with common modifiable causes including inadequate preoperative assessment, suboptimal scheduling, and insufficient postoperative bed availability (24, 25). Cancellation reduction therefore represents a high-yield target for efficiency improvement.

Quality, by contrast, is a multidimensional concept described in the literature across multiple domains. Quality measures may be grouped into structure, process, and outcome indicators, thereby capturing the organisational conditions for care, the care delivered, and its results (26). It encompasses not only efficiency, which, as noted above, is closely related to resource optimisation, but also effectiveness, that is, the extent to which care achieves the desired clinical outcome. In addition, quality also includes patient safety and patient-centredness, with attention to minimising the risk of harm and adverse events and to engaging patients in decisions about their care. As noted above, timeliness is also important, and quality assessment further places emphasis on equity and access to care, i.e. ensuring fair and just access to care as well as the availability of services (27). More recently, quality assessment has also increasingly incorporated composite measures, such as textbook outcome, which combine several desirable perioperative outcomes into a single metric of optimal care (28). In short, it concerns whether surgical care is safe, appropriate, effective, and patient-centred, and whether it achieves desirable outcomes for patients.

According to earlier findings, improving the efficiency of the operating room can be achieved through several measures, highlighting several strategies to enhance the efficiency of operating rooms. The first step is to identify the root causes and investigate the reasons behind inefficiency and address them through a multidisciplinary approach. This approach should involve awareness training and personal accountability for all staff involved in the process (29). To be able to better facilitate this, accurate data collection and analysis can provide valuable insights into areas for

improvement. Consequently, perioperative organisations need to strive for the standardisation of tasks and to maintain effective communication among the entire surgical team (30). In addition, interpersonal competencies can be important for effective surgical team functioning (31). At the same time, productivity in surgery should not be pursued as an end in itself, as efficiency gains that are not carefully considered may come at the expense of both the work environment and patient safety.

By adopting these strategies, hospitals and healthcare systems can enhance the productivity of their operating rooms while optimising their limited resources. But it is also important to remember that successful surgical outcomes depend on collaborative efforts, effective communication, and a focus on patient well-being. However, there are various challenges that can negatively impact the effectiveness of surgical teams. This broader interpretation is consistent with qualitative work showing that members of surgical teams understand operating room efficiency as extending beyond production alone to include team functioning and the delivery of value and quality (32).

These concepts are closely related but not interchangeable. Productivity primarily concerns output, efficiency concerns the relationship between output and the time and resources used to achieve it, throughput reflects the rate of patient flow through the system, and capacity refers to the resources available to deliver care. Quality, by contrast, concerns whether care is delivered safely, effectively, and in a patient-centred manner. Thus, high output alone should not be equated with high performance, as surgical care must also be efficient, sustainable, and safe. The distinctions between these concepts, and their relevance to the individual studies, are summarised in Table 1.

Table 1. Definitions, typical indicators, and relevance of key concepts used to describe surgical performance in this thesis.

Concept	Core meaning in this thesis	Typical indicators	Relevance to thesis
Productivity	Surgical output	Cases per OR/day	Study I
Efficiency	Output relative to time/resources	TCT, NOT, utilisation	Studies I–II
Throughput	Patient flow through pathway	Waiting time, completed cases	Studies II–IV
Capacity	Available resources to deliver care	OR time, staff, beds	Studies II–III
Quality	Safe, timely, patient-centred care	Outcomes, PROMs, safety	Studies III–IV

Note. OR, operating room; TCT, total case time; NOT, non-operative time; PROMs, patient-reported outcome measures.

Donabedian, who described healthcare quality in terms of structure, process, and outcome, also identified seven attributes of healthcare quality: efficacy, effectiveness, efficiency, optimality, acceptability, legitimacy, and equity. These attributes complement, but are conceptually distinct from, the Structure–Process–Outcome model. Because the present thesis primarily examines how organisational conditions relate to care processes and outcomes, Structure–Process–Outcome is used as the principal organising framework, while the seven attributes are not operationalised separately (33).

Organisational conditions and staff experience in perioperative work

An efficient and safe surgical operation requires a well-organised and coordinated operating department. The organisation must be able to manage a range of factors such as staff availability, equipment, and materials, as well as patients' needs. At the same time, hospitals face

constant pressure to increase productivity and reduce costs, which can lead to a risk of negative effects on patient safety and the quality of care. These productivity pressures have direct organisational consequences, contributing to staffing shortages, increased workload demands, and reduced capacity for adaptive coordination at the frontline (34). There is also a growing trend of introducing different kinds of productivity enhancing paradigms, such as Lean management or Six Sigma in healthcare, all of which aim to increase efficiency and reduce waste within organisations (35, 36). A study from 2021 showed that the introduction of Lean Management in an operating department led to a reduction in waiting times and an increase in patient volumes, while improving the quality of care and staff job satisfaction (37). Effective organisation of operating departments has been shown to reduce operating costs, shorten hospital stays, and decrease postoperative morbidity, while poorly organised teams conversely increase the risk of complications and longer length of stay (38).

Operating rooms are inherently resource-scarce environments, with high demand for limited physical, human, and time-based resources creating both operational and ethical challenges. OR time itself is a scarce and valuable commodity, constrained by finite daily hours and growing surgical demand, particularly in the context of ageing populations and rising healthcare utilisation (39). Resource constraints and staffing shortages therefore represent persistent organisational challenges in perioperative settings. Insufficient staffing not only limits OR capacity directly but also places disproportionate demands on available personnel, increasing the risk of fatigue, errors, and staff turnover (34). Leadership plays a central role in mediating these pressures as effective perioperative leadership has been associated with better team coordination, clearer role definition, and more sustainable workload distribution (38). At the same time, perioperative organisations frequently depend on informal adaptations by frontline staff to maintain function under resource-constrained conditions. Such adaptive behaviour, while often invisible in formal performance metrics, is essential to day-to-day operational continuity and represents a form of organisational resilience that formal efficiency initiatives risk eroding if not carefully accounted for (40). Organisational conditions thus shape not only what care is delivered, but how work is experienced by those delivering it. The interplay between structural factors: staffing levels, scheduling practices, physical layout, and available resources, and staff experience has implications for both performance and sustainability (41).

One established instrument for assessing staff attitudes to safety and organisational conditions in the operating room is the Safety Attitudes Questionnaire–Operating Room version (SAQ-OR). The Swedish version contains 57 items, of which 30 contribute to six domains: teamwork climate, safety climate, job satisfaction, stress recognition, perceptions of management, and working conditions. Domain scores are commonly transformed to a scale from 0 to 100, with higher scores indicating more positive attitudes. The instrument has been translated, culturally adapted, and psychometrically evaluated in Swedish operating-room settings, with subsequent evaluation supporting its use while also showing some variation in psychometric performance across professional groups (42, 43).

Breast cancer surgery as a clinical and organisational context

Breast cancer is a disease characterised by the uncontrolled growth of malignant cells in the breast. It is the most common type of cancer among women worldwide (44). In 2024, about 8,800 women were diagnosed with and 1,328 died from breast cancer in Sweden, making it the most common malignancy among women in the country (45).

The aetiology of breast cancer is multifactorial, with both genetic and environmental factors playing a role. The incidence of breast cancer in Sweden has increased over the years, from about 90 women per 100,000 in 1970 to over 150 women per 100,000 per year in 2019 (46). However, the mortality rate has slowly declined, indicating improvements in detection and treatment strategies (44, 46).

Surgery is a common treatment option for breast cancer and the basis for curative treatment. Breast cancer care is delivered within a multidisciplinary framework in which surgery, oncological treatment, and radiotherapy constitute the three core treatment modalities (47, 48). The type of surgery depends on the stage and type of the cancer, the size of the tumour in relation to the size of the breast, the patient's overall health, and patient preferences (49). In Sweden, the surgical procedures for both in situ cancer and invasive breast cancer include mastectomy with or without reconstruction, breast conserving surgery (BCS)/lumpectomy with irradiation, and less commonly BCS/lumpectomy without irradiation (50). In addition, lymph node staging is performed by removing the first lymph node(s) draining the cancer i.e. the sentinel lymph node biopsy (SLNB) or if extensive axillary metastasis also by axillary lymph node dissection (ALND).

Mastectomy involves the removal of the entire breast, while BCS/lumpectomy involves the removal of the cancer and some surrounding normal tissue. Breast-conserving surgery is commonly followed by radiotherapy to reduce the risk of local recurrence (49). When neoadjuvant systemic therapy is administered prior to surgery, the aim is typically to treat potential micrometastatic disease and optimise surgical conditions rather than to replace the surgical procedure itself, thus underscoring the role of surgery as the definitive local treatment (47). Surgical techniques have developed over time, and oncoplastic approaches have expanded the possibilities for breast-conserving, tissue-sparing surgery (51). As shown in Figure 3, breast-conserving surgery is the dominant approach among women with small invasive breast cancers in Sweden, illustrating the relative procedural consistency of this clinical context (52).

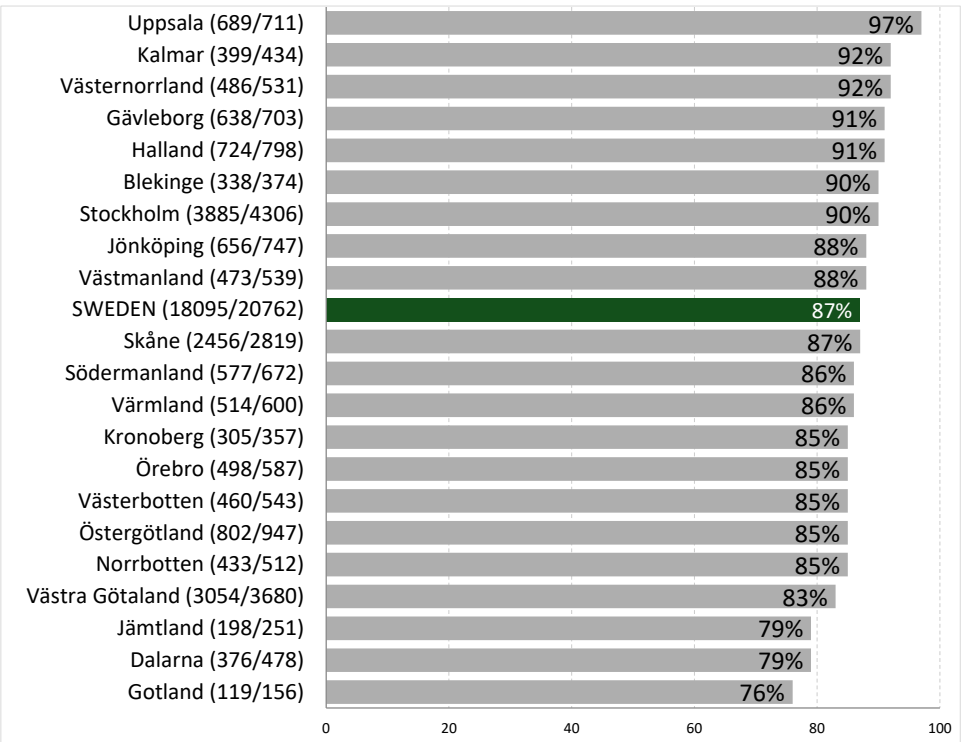


Figure 3. Proportion of women with small invasive breast cancers (<3 cm) who underwent breast-conserving surgery in Sweden between 2020 and 2024, by region. Numbers in parentheses indicate the number of women undergoing breast-conserving surgery/total number of eligible cases. Region was missing for 17 cases, of whom 15 underwent breast-conserving surgery; these cases are included in the national total but are not shown separately. Data from the NKBC interactive report (52).

Timely access to surgery is recognised as a key quality dimension in breast cancer care. In Sweden, breast cancer care is organised within the national standardised care pathway framework, *standardiserat vårdförlopp* (SVF), introduced in 2015 to improve coordination and reduce avoidable waiting and uncertainty. For breast cancer, national targets specify the interval within which treatment should commence. Regardless of where in the country they seek care, the current national target stipulates that at least 80% of patients should commence treatment within 35 calendar days of a well-founded suspicion of cancer (53). National registry data also show that the median lead time from sampling to primary surgery, which had remained around 28 days through 2021, increased to 34 days in 2023 (54). Across all cancer diagnoses combined, the national rate of patients meeting SVF time targets declined from 42% in 2022 to 40% in 2023, the lowest level since SVF was introduced, with the shortfall most pronounced for surgical treatment pathways (55).

The waiting period between diagnosis and surgery carries not only organisational but also psychological significance. The period between diagnosis and surgery can be associated with substantial psychological distress, including fear, uncertainty, and concerns about the future (56). Furthermore, a systematic review and meta-analysis demonstrated a significant reduction in both depression and anxiety following breast cancer surgery, suggesting that the surgical intervention itself has a psychologically relieving effect and that delays to its completion are not merely logistical but clinically meaningful for patient-reported outcomes (57).

Breast cancer thus represents a particularly suitable context for studying the intersection of surgical organisation, waiting times, and patient-reported outcomes. Its high incidence generates sufficient case volumes for quantitative registry-based research. The standardised care pathway defines a nationally consistent treatment process with explicit, measurable time points, which makes it possible to examine organisational factors such as waiting time while limiting variation in clinical management as a source of confounding. The surgical procedures involved, primarily mastectomy and breast-conserving surgery with sentinel node biopsy, form a well-defined and relatively homogeneous set of interventions, further reducing procedural heterogeneity as an analytical obstacle. The national quality register underpinning this research, and its collection of patient-reported outcomes, is described in the following section.

Quality registers, NKBC and PROMs

National quality registers occupy a central place in the Swedish healthcare and research infrastructure. Sweden has developed more than one hundred national healthcare quality registers, many of which contain detailed clinical data that complement information available in mandatory health registers and administrative databases. Their value for research lies not only in the amount of data collected, but in the combination of clinical detail, prospective registration, population-based coverage, and the possibility of follow-up over time (58). In Sweden, this infrastructure is further strengthened by the personal identity number, which enables individual-level linkage between data sources and has become a central methodological foundation for register-based medical research (59). At the same time, register data should not be treated as uniformly complete or valid by default. Coverage, completeness, validity, and timeliness vary between registers and variables, and each register-based study therefore requires attention to data quality, definitions, and patterns of missingness (58).

Increasingly, this register infrastructure has been extended to include the patient perspective through patient-reported outcome measures (PROMs). PROMs capture patients' own assessments of their health, functioning, symptoms, and wellbeing, rather than relying on clinician-rated outcomes. In cancer care, PROMs are used to capture dimensions of health and wellbeing that may be poorly reflected in clinician-rated endpoints, including symptom burden, functioning, health-related quality of life, and psychological wellbeing (60). This is particularly relevant in surgical oncology, where traditional outcomes such as survival, complications, and recurrence do not fully capture the psychological burden of diagnosis, waiting, treatment, and recovery. Integrating PROMs into cancer care and research can therefore broaden quality assessment from clinical and administrative endpoints to outcomes that are directly meaningful to patients (61).

The European Organisation for Research and Treatment of Cancer (EORTC) quality-of-life measurement system has a modular structure in which the cancer-generic Quality of Life Questionnaire Core 30 (QLQ-C30) is combined with diagnosis-specific modules, such as the breast cancer module Quality of Life Questionnaire Breast Cancer 23 (QLQ-BR23). The QLQ-C30 contains 30 items covering functional domains, symptom domains, and global health status and quality of life, while QLQ-BR23 adds breast cancer-specific aspects of functioning and symptom burden. Multi-item scale scores are conventionally transformed to a range from 0 to 100 (62). Higher scores indicate better functioning or global

quality of life on the functional scales, but greater symptom burden on the symptom scales. There is no single universal threshold defining a "good" PROM score; interpretation may instead be based on changes over time, differences between groups, or comparisons with clinically relevant reference populations (63–65).

The Swedish National Quality Register for Breast Cancer (NKBC; In Swedish: *Nationellt Kvalitetsregister för Bröstcancer*) brings these two elements together. Established in 2008 and maintained within the national cancer register infrastructure, NKBC records information on lead times, diagnostics, tumour characteristics, preoperative oncological treatment, breast and axillary surgery, postoperative oncological treatment, and follow-up (66). Because NKBC is assessed in relation to the legally mandated Swedish Cancer Register, case coverage is high. Validation has shown coverage approaching 100%, with good completeness, comparability, and agreement for core clinical variables against medical records, although improved timeliness has been identified as important for clinical quality monitoring (67).

Since 2020, NKBC has also collected PROMs prospectively. This extends the register from clinical and process indicators to patients' own assessments of health and wellbeing, while also requiring a distinction between the near-complete coverage of core clinical NKBC data and the more limited and gradually increasing coverage of PROM data (66). For the present thesis, this combination is important because NKBC makes it possible to link an organisational process-level variable, diagnosis-to-surgery lead time, with patient-reported psychological outcomes within the national register setting. In this way, NKBC and its PROM component provide an empirical link between surgical organisation, timely access to care, and patient-reported outcomes.

Waiting time, access to care and patient outcomes

Timing of breast cancer surgery is increasingly recognised as an important quality indicator. Recent work notes that the Commission on Cancer introduced a 60-day quality metric for surgery after diagnosis in patients not receiving neoadjuvant treatment, and subtype-specific analyses suggest that prolonged time to surgery may be particularly harmful in HR+/HER2– disease (68). Earlier observational research suggests that longer delays between diagnosis and surgery may adversely affect long-term survival in breast cancer, with Swedish population-based data showing an association between longer time to surgery and poorer overall survival (69). Breast cancer surgical delays appear to arise through an interaction between diagnostic complexity, organisational design, and

coordination across the pathway. In a national English analysis of 236,205 patients, shorter waiting times were associated with straight-to-test appointments, multiple tests on the same day, and discussion in a single multidisciplinary team meeting, whereas cancelled appointments and attendance at multiple providers prolonged the pathway (70).

In the Swedish context, this issue is reflected in the proportion of patients who undergo surgery within the SVF-specified target interval. As described above, the SVF framework defines explicit time targets for the interval between a well-founded suspicion of cancer and start of treatment (53). Figure 4 shows that the proportion of patients undergoing breast cancer surgery within this target interval declined from 49% in 2020 to 29% in 2023. From 2024 onward, the target interval was extended from 28 to 35 calendar days, after which the proportion increased to approximately 47–49%. This apparent recovery should therefore be interpreted with caution, as it partly reflects a change in the specified target interval rather than necessarily a corresponding improvement in actual access to surgery. Nevertheless, the consistently low proportion of patients treated within the SVF target illustrates that timely access to breast cancer surgery remains a persistent organisational challenge.

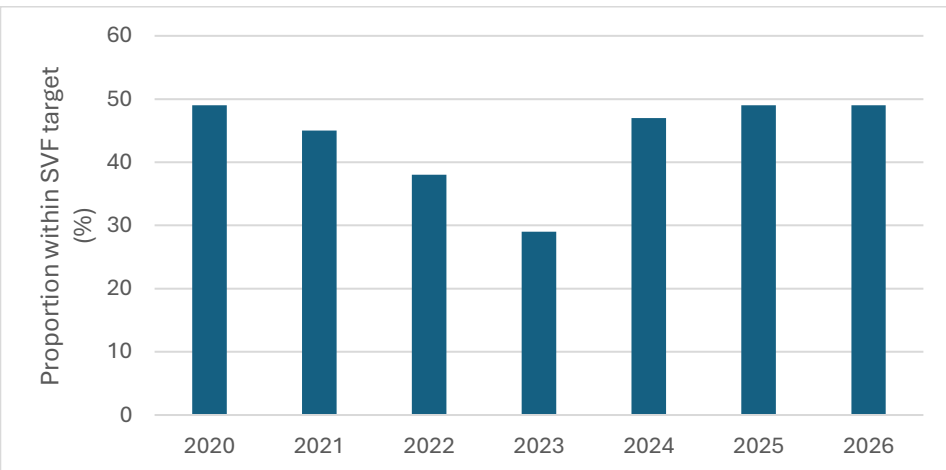


Figure 4. Proportion of patients undergoing surgery within the SVF-specified target interval, 2020–2026. Data for 2026 were retrieved on 8 June 2026. The target interval changed in 2024 from 28 to 35 calendar days. Data from (71).

These measures describe different intervals and populations: the 34-day median above refers to sampling-to-surgery time in national NKBC data, Figure 4 shows the SVF interval from well-founded suspicion to start of

treatment using the target applicable in each year, whereas Study IV measures histopathological diagnosis-to-surgery lead time within the selected PROM cohort.

This suggests that timely access is relevant both clinically and organisationally in breast cancer care. Waiting time can also be interpreted as a systems-level measure of how effectively capacity and coordination translate into access to care. It may serve as a proxy for how well structural resources are translated into timely care. Regardless of its impact on tumour biology, the waiting period may affect patients' psychological wellbeing and their experience of care, justifying timeliness as a quality objective in its own right, independent of its effect on clinical endpoints. Delays may arise from structural constraints, variations in coordination, and differences in clinical processes, while also influencing both patient and system-level outcomes. To analyse these interrelated dimensions across the four studies, this thesis draws on the Donabedian model of healthcare quality (72), which is outlined in the theoretical and analytical framework below.

Knowledge gap and rationale for the thesis

Research on surgical organisation and perioperative care has grown substantially over the past two decades, generating important insights into operating room efficiency, staff workload, waiting times, cost containment, and patient outcomes. However, this body of literature has largely developed in parallel streams, with studies tending to examine one dimension of performance in isolation. Efficiency research focuses on throughput and resource utilisation (38, 73); workforce studies examine staff experience and retention (34); and outcomes research tracks clinical or patient-reported endpoints (60, 61, 64). Rarely are these perspectives brought together within a single analytical framework applied to the same organisational context.

The Swedish context illustrates this fragmentation particularly clearly. Surgical care is organised across 21 administratively autonomous regions with considerable variation in capacity, staffing, and performance. Long waiting times have been a persisting problem in Swedish healthcare, with substantial differences in waiting times between regions and between different parts of the healthcare system. This has prompted a series of national policy initiatives including the SVF system and, most recently, a government inquiry into a strengthened waiting time guarantee (74). The Swedish Perioperative Register, launched in 2011, has created a national database of perioperative cases and time stamps, providing an

infrastructure for monitoring variation in surgical capacity across the country (75). Although these developments have improved the monitoring of waiting times and perioperative performance, they have not resolved the underlying analytical challenge of understanding how organisational differences translate into variation in care processes and patient outcomes. Despite this infrastructure, research linking organisational variation in perioperative care to downstream patient-level outcomes remains limited, and the mechanisms through which such differences may be linked to waiting times and patient experience are still poorly understood.

This fragmentation has practical consequences. A surgical department may improve narrow OR utilisation metrics while overlooking how delays and workforce pressures propagate through the system. Surgeon exposure to turnover delays has been associated with increased adverse events, while perioperative nursing shortages may undermine sustained capacity and workforce stability (9, 34). Without an integrated perspective that traces the connections between organisational structure, care processes, and outcomes at both system and patient level, it is difficult to understand how organisational decisions propagate through the system, or to identify where interventions are most likely to have lasting effects.

This problem is compounded by a structural tension between short-term production optimisation and long-term organisational sustainability. Pressure to maximise throughput and reduce costs can, if poorly managed, erode the very conditions that make sustained performance possible. Perioperative workforce data illustrate this tension, indicating a marked deterioration in nurse staffing over the past decade, with reported vacancy rates rising substantially and many managers describing moderate to critical shortages (34). Although these figures are not Sweden-specific, they highlight a broader perioperative workforce instability that is relevant to surgical capacity. Such shortages may contribute to a self-reinforcing cycle in which reduced staffing increases workload, lowers job satisfaction, and accelerates turnover, thereby weakening the organisational capacity needed for sustained productivity.

There is therefore a need for research that examines perioperative organisation as a coherent whole rather than a collection of separable components. Systematic reviews of OR organisation and performance have identified this gap, noting that studies tend to focus on narrow performance metrics without integrating staff, patient, and system-level perspectives (38, 73). In particular, the link between how surgical departments are organised and what patients ultimately experience,

including not only clinical outcomes but also patient-reported dimensions such as worry and psychological wellbeing, remains underexplored. Evidence concerning the effects of organisational conditions on patient-reported outcomes is lacking even in areas where it might be expected, such as interprofessional collaboration (76). Waiting time provides a particularly illuminating lens for this inquiry, as it simultaneously reflects organisational capacity, coordination quality, and pathway design, while also carrying documented consequences for patients' experience of care.

This thesis addresses these gaps through four interrelated studies, each positioned at a distinct level of the Donabedian Structure–Process–Outcome framework (72, 77). The four studies address different parts of a multilevel relationship between organisational form, care processes, system-level outcomes, and patient experience.

Theoretical and analytical framework

To provide a coherent analytical framework for the four studies, this thesis draws on the Donabedian model of healthcare quality, which distinguishes between Structure, Process, and Outcome (72, 77). The model was selected because it provides a well-validated and widely applied conceptual language for analysing how organisational conditions shape care delivery and patient outcomes, precisely the chain of relationships this thesis seeks to examine. Unlike more mechanism-focused theories, the Donabedian model is broad enough to accommodate studies that vary in design and level of analysis, from objective productivity measures to patient-reported outcomes, while maintaining a coherent analytical logic across all four studies. The model is used here as a heuristic and organising framework, not as a strict classification system, to articulate the relationships between organisational conditions, care processes, and different types of outcomes in the surgical setting.

Structure refers to the organisational conditions under which surgical care is delivered, including staffing arrangements, skill mix, resource allocation, and scheduling practices. Study I operationalises these structural differences by comparing two differently organised surgical departments performing identical procedures, thereby reducing variation related to surgeon and procedure while preserving the organisational contrast.

Process refers to what actually occurs in the delivery of care, the workflows, interactions, and practices that constitute everyday surgical work. Study I captures process through objective productivity measures

such as total case time and non-operative time. Study III deepens this by examining how organisational conditions are experienced and enacted by staff in their everyday perioperative work, addressing the mechanisms that connect Structure to Process. Within Study III, Self-Determination Theory is used to interpret how specific organisational routines support or thwart staff needs for autonomy, competence, and relatedness (78).

Outcome is understood at two levels. At the system level, Study II examines cost-efficiency and organisational capacity as intermediate outcomes, demonstrating how structural differences translate into differences in resource stewardship and waiting times. At the patient level, Study IV examines the association between a process-level measure of timely access, diagnosis-to-surgery lead time, and patient-reported worry about future health. Lead time may reflect underlying organisational capacity and coordination, but Study IV does not directly identify the structural determinants of the observed waiting times.

Within this framework, the four studies address organisational structure, care processes, and outcomes at different analytical levels. Figure 5 summarises how the four studies are situated within the Donabedian framework used in this thesis.

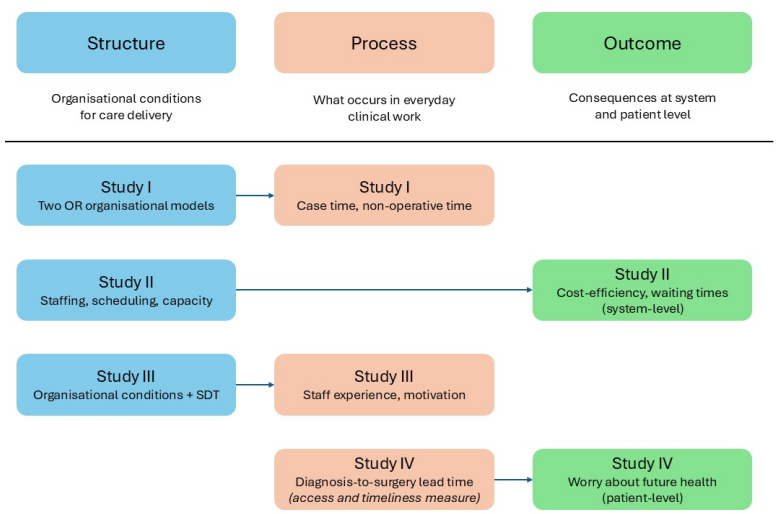


Figure 5. The Donabedian Structure–Process–Outcome model as an organising framework for the four studies of this thesis. Structure refers to the organisational conditions under which care is delivered; Process to what occurs in everyday clinical work; and Outcome to consequences at both system and patient level. Arrows indicate the analytical direction from organisational conditions through care processes to outcomes. Studies I and III span Structure and Process. Study II links

structural conditions to system-level outcomes, whereas Study IV links a process-level measure of access and timeliness to a patient-level outcome. The figure represents a thesis-level conceptual synthesis and should not be interpreted as a single empirically tested causal chain.

Aims of the thesis

The overall aim of this thesis is to examine how differences in surgical department organisation relate to perioperative productivity, resource utilisation, staff experiences, and patient-reported outcomes, and to explore the consequences of organisational structure across multiple levels of the healthcare system.

The specific aims of each of the four studies are:

I. To investigate how total case time differs between two differently organised surgical departments, with the same set of surgeons operating in these two departments.
II. To compare the health-economic consequences of two organisational models for breast cancer surgery, including differences in perioperative costs, capacity, and waiting time implications.
III. To explore how operating room and anaesthesia nurses experience productivity, workflow, and organisational conditions in their work environment, and how these relate to autonomy, competence, and relatedness.
IV. To examine the association between waiting time to breast cancer surgery and patient-reported worry about future health, as well as other patient-reported outcomes.

Materials and Methods

This thesis consists of four interrelated studies using quantitative, qualitative, and mixed-methods approaches to examine how surgical department organisation is related to perioperative productivity, staff experience, resource use, and patient-reported outcomes. The four studies were conducted in Swedish surgical settings using perioperative registry data, operational and financial data, survey responses, interviews, and national breast cancer registry data.

Setting and ethical considerations

Studies I–III were conducted in a Swedish university hospital setting, whereas Study IV was based on data from a Swedish national quality

register. Studies I–III were covered by ethical approval from the Regional Ethical Review Board in Umeå, Sweden (reg. no. 2017/463-31). Study IV was additionally covered by an amendment approved by the Swedish Ethical Review Authority (reg. no. 2025-06241-02).

In Studies I and II, registry and medical record data were handled confidentially and pseudonymised during data collection. In Study III, the survey data were collected using participant-held codes. The researcher could not link codes to individual participants, but participants could request withdrawal of their response by providing their code. All participants in the qualitative component provided written informed consent prior to participation. For the retrospective and registry-based studies, consent was managed in accordance with the ethical approvals and applicable Swedish regulations. In Study IV, the data were anonymised before the authors were granted access. All data were de-identified for analysis, and where code sheets or re-identification keys were used, access was restricted and keys were stored separately. Data were handled in accordance with applicable data protection regulations, including GDPR.

Because the main author knew, and was known to, participants in the interviews in Study III, this entailed both ethical and methodological sensitivity. To reduce the risk of dependency or perceived pressure, voluntariness, participant information, and confidentiality were emphasised throughout the recruitment and interview process.

Overview of the four studies

The four studies included in this thesis were designed to address complementary aspects of the overall aim. The studies combine quantitative, qualitative, and mixed-methods approaches and draw on perioperative and organisational records, financial and registry data, staff surveys, and individual interviews.

Each study builds on, or extends, insights from the others in order to provide a broader understanding of how surgical organisation shapes productivity, resource use, staff experience, and patient-reported outcomes related to waiting time. The sequence moves from the local perioperative organisation to health-economic and staff perspectives, and finally to the patient perspective.

Study I examined how two different organisational models were associated with differences in surgical productivity. Study II extended this

work by investigating the health-economic consequences of these two models and by simulating their implications for waiting time within the studied patient cohort. Study III then explored how operating room and anaesthesia nurses experienced their organisational work environment and its relation to everyday productivity. Finally, Study IV examined patient-reported worry about future health, as well as other patient-reported outcomes, in relation to waiting time for breast cancer surgery.

Study I

Design and study population

Study I was a retrospective observational study of patients undergoing elective surgery under general anaesthesia for breast cancer or malignant melanoma at a Swedish university hospital between 2012 and 2016. Because elective surgical volume was considerably higher at the general surgery (GS) department, cases from that department were included only from January to May each year, whereas cardiothoracic (CT) cases were included across the full 2012–2016 period, in order to keep the two departmental cohorts comparable in size. The study compared two surgical departments with different organisational conditions and patterns of perioperative productivity: a general surgery department and a cardiothoracic surgery department. The departments differed in several organisational characteristics, including anaesthesia staffing, nursing skill mix, team working hours, and access to auxiliary personnel, while the same group of surgeons performed comparable procedures in both settings. This reduced variation related to surgeon while preserving the organisational contrast between departments. Patients were allocated from a shared waiting list, primarily according to order of referral rather than patient characteristics, although practical scheduling considerations, such as expected procedure duration, could also influence allocation. Patients were identified and classified using diagnostic codes, procedure type, and intervention codes from the Swedish Classification of Health Care Measures (KVÅ) (79). The study population was restricted to elective cases with the intention of increasing comparability between departments and reducing the influence of acute logistical variation.

Data collection

Data were extracted from Orbit, the hospital's electronic perioperative planning and scheduling system, and were supplemented with information from perioperative records, surgical reports, and departmental staffing data. The collected material included patient

characteristics, diagnosis, type of procedure, perioperative time variables, and staffing-related information relevant to the comparison between the two departments. Data were also collected on organisational and surgeon-related factors, such as the presence of a surgical trainee or assistant surgeon, whether the operating surgeon was a consultant, and the surgeon's level of experience. These variables were included to better describe the organisational context in which each case was performed. The study period was chosen to maintain a relatively stable organisational context, including a consistent group of surgeons and staff at the two departments, thereby reducing the influence of broader organisational changes over time.

Potential inaccuracies in Orbit time stamps were addressed by excluding cases with clear registration errors, and the remaining registrations were considered reasonably reliable due to their temporal proximity to the perioperative events.

Analysis

The primary outcome was total case time, measured in minutes and defined as the time from room set-up start until room clean-up finish. This outcome was chosen to capture the overall perioperative time used for each case and thereby reflect differences in workflow efficiency between the two departments. Descriptive statistics were used to characterise the study population and the organisational context of the two departments. Baseline differences between the departments, including differences in organisational conditions and case characteristics, were examined using chi-square tests. Linear regression was then used to estimate the association between department and total case time. The model was adjusted for clinically relevant covariates measured prior to allocation to department, including age, sex, BMI, American Society of Anesthesiologists physical status classification (ASA) score, surgeon experience, consultant status, and the presence of a surgical trainee or assistant surgeon. These covariates were selected in order to account for relevant patient- and organisational factors that could influence perioperative time use independently of department. Missing values were handled by listwise deletion. Regression model assumptions were assessed using standard diagnostic procedures, including evaluation of residual distribution. Additional analyses were performed to assess the robustness of the findings in relation to model specification and covariate adjustment. All statistical analyses were performed in R version 4.5.0.

Study II

Design and study population

Study II was a health-economic evaluation using a bottom-up micro-costing and cost-minimisation approach to compare breast cancer surgery performed at the two surgical departments examined in Study I. The study built directly on the organisational comparison established in Study I and assessed whether the observed differences in perioperative productivity were also associated with differences in costs, efficiency, and waiting time implications. Because the two departments operated within the same hospital and provided the same surgical treatment to a comparable patient group, the analysis assumed equivalent clinical outcomes and therefore focused on differences in resource use and perioperative costs. Data from Study I were supplemented with financial and resource-use data. To create a more homogeneous study population, the analysis was restricted to patients undergoing breast cancer surgery. The Study II cohort therefore overlaps with, but is not nested within, the Study I cohort. Study II excluded malignant melanoma and was limited to January 2014–September 2016, but, unlike Study I, it included general surgery cases from all calendar months (not only January–May) and retained cases with missing BMI, which was not required as a covariate. The full-year general surgery inclusion is what makes Study II larger overall (543 vs 526) despite its narrower diagnostic scope, while the shorter inclusion period reduced the cardiothoracic count (98 vs 177). The evaluation was conducted in the same Swedish university hospital setting as Study I and was restricted to the perioperative phase of care.

Data collection

Data were collected from perioperative records, operating room logs, and hospital administrative and financial systems. The study population consisted of 543 consecutive patients undergoing elective breast cancer surgery at the two departments included in Study I. These data were used to identify case-level resource use during the perioperative phase of care.

Detailed information was obtained on operating room time and the use of personnel resources during the different phases of the perioperative process. Data on staffing composition, occupational categories, and time use were combined with salary information to estimate personnel-related costs at patient level. In addition, data on consumables, medications, and other direct perioperative resources were collected and assigned to the included surgical cases.

Facility-related costs were also collected, including the cost of operating room use. Financial records and local administrative data were used to assign costs to the identified resources within the hospital setting. The costing was based on observed resource use within the study dataset rather than on external benchmark values, to reflect the organisational conditions of the two departments as closely as possible.

Resource use was measured from 2014–2016 operating logs and direct observations, while unit costs were valued using 2025 price data from regional payroll, pharmaceutical, and facility systems. This replacement costing approach reflects the current cost level of the identified resources without requiring additional inflation adjustment or indexation assumptions, as a single price year was applied consistently across all cost components.

Analysis

The analysis was conducted from a healthcare provider perspective, as the aim was to examine differences in perioperative resource use and organisational efficiency within the hospital setting. A cost-minimisation approach was used because the two departments were assumed to provide equivalent clinical outcomes, and the analysis therefore focused on differences in perioperative costs and resource use between the general surgery department and the cardiothoracic surgery department. To reduce unnecessary variation in the cost comparison, hospital-level overhead and certain non-differential costs were assumed to be equal between the two departments and were therefore not included as primary cost drivers in the analysis. However, operating room rent was included, as this was considered relevant for the generalisability of the findings.

Cost data were summarised using means, medians, and standard deviations. Uncertainty in total and component-level cost differences was quantified using a grouped, case-level non-parametric bootstrap, in which cases were resampled within department while preserving group sizes. A total of 1,000 replications were performed, and percentile-based 95% confidence intervals were calculated. The primary contrast was defined as the mean cost difference between departments (GS minus CT).

In addition to the direct cost comparison, a deterministic simulation model was used to estimate the implications of departmental differences in procedural efficiency for operating room capacity and preoperative waiting time. The simulation was based on observed patient flow and mean operation times from the study dataset, assuming steady weekly

inflow and constant operating room availability. Capacity was derived from mean total case time, and waiting time scenarios compared a baseline situation with CT capacity available ("CT on") to a scenario where CT capacity was removed ("CT off"). Sensitivity analyses were also performed by varying anaesthesiologist presence (0%, 40%, 50%, and 100%) in order to assess the robustness of the findings. Simulation has previously been used to examine maximum surgical-suite capacity while accounting for the balance between resource utilisation, patient readiness, and waiting (80). All statistical analyses were performed in R version 4.5.0. The reporting of the health-economic evaluation was informed by the Consolidated Health Economic Evaluation Reporting Standards (CHEERS) 2022 statement (81).

Study III

Design and study population

Study III used a sequential explanatory mixed-methods design consisting of a quantitative survey phase followed by a qualitative interview phase. The questionnaire results were used as contextual signals to inform the focus and sampling of the interviews rather than as a performance assessment of the two included departments. The study was conducted at the same two operating room departments as in Studies I and II, here denoted as Unit A and Unit B.

All operating room nurses, anaesthesia nurses, and nursing assistants at the two departments were invited to participate in the survey. A total of 47 questionnaires were returned (n=47; Unit A=28, Unit B=19). In the second phase, maximum-variation sampling was used among operating room nurses and anaesthesia nurses for the purpose of capturing variation in unit affiliation and level of experience. The interview phase included 24 participants (n=24; Unit A=8, Unit B=16). Inclusion criteria for the interview phase were at least 1 year of experience in the professional role and at least 6 months at the current department, in order to ensure sufficient familiarity with unit workflow and organisation. Among interview participants, years of OR experience ranged from 3 to 26 years (median 9). Participants were recruited through staff meetings, workplace information, and direct approach on site by a member of the research team. Interested participants contacted the researchers by telephone or email. No incentives were offered.

Data collection

The first phase used the Swedish version of the Safety Attitudes Questionnaire–Operating Room (SAQ-OR), which has undergone translation, cultural adaptation, and psychometric evaluation in Swedish operating-room settings (42, 43). Data collection was conducted in October 2022. The questionnaire covers six domains: safety climate, teamwork climate, job satisfaction, working conditions, stress recognition, and perceptions of management. In the second phase, one-on-one semi-structured interviews were conducted in Swedish by the first author in two rounds. The first round comprised 16 interviews at Unit B in December 2022, and the second round comprised 8 interviews at Unit A between January and March 2025. Interviews were conducted in private rooms on site, audio-recorded, and transcribed verbatim. The first two interviews included a non-managerial observer for procedural credibility. Brief field notes were made during and immediately after the interviews. Interview duration ranged from 38 to 75 minutes (median 61 minutes). The interview guide focused on experiences of productivity, planning and workflow, staffing, collaboration, learning, and perceived enablers and barriers in everyday perioperative work, and was refined iteratively during the data collection process.

Analysis

In the first phase, survey data from the SAQ-OR were analysed across its six domains: safety climate, teamwork climate, job satisfaction, working conditions, stress recognition, perceptions of management. Domain scores were dichotomised, and proportions between the two units were compared using Fisher's exact test. All statistical analyses were performed in R version 4.5.1.

The interview material was analysed using qualitative content analysis with an inductive approach. Coding proceeded through a manifest hierarchy of subcategories, categories, and main categories. Initial coding was kept close to participants' descriptions of everyday perioperative work, while category development involved progressive abstraction and repeated comparison with the underlying data. The analysis was guided by established principles for achieving trustworthiness in qualitative content analysis (82, 83).

Self-Determination Theory was introduced in a subsequent abductive interpretive phase, rather than as a deductive framework from the outset. In this phase, the empirically derived categories were examined through the lens of autonomy, competence, and relatedness in order to explore how organisational conditions supported or thwarted sustainable

productivity in everyday work. This analytical sequence made it possible to preserve the data-near character of the initial analysis while also relating the findings to a broader theoretical framework.
Integration occurred through connecting and building: survey findings informed interview sampling and the focus of the interview guide, while the quantitative and qualitative findings were brought together during interpretation within a common explanatory account (84).
To strengthen the credibility of the analysis, one transcript was independently double-coded to calibrate the approach, and interim codebooks and category proposals were discussed with co-authors from different professional backgrounds. The analysis was thus refined through repeated discussion and adjudication rather than by relying solely on a single researcher's interpretation.
Because the first author was clinically familiar with the study setting, the interviews and subsequent interpretation were deliberately kept mechanism-focused rather than evaluative.

Study IV

Design and study population

Study IV was a longitudinal cohort study examining associations between diagnosis-to-surgery lead time and patient-reported outcomes in women with primary invasive breast cancer. The study used data from NKBC, a population-based national quality register with near-complete coverage of incident breast cancer in Sweden. Eligible patients were women aged 18 years or older with confirmed primary invasive breast cancer who underwent primary breast surgery during the period January 2020 to December 2024 and had at least one patient-reported outcome measure observation. Patients were excluded if they had a prior breast cancer diagnosis, if the recorded lead time was less than one day or exceeded 365 days, if data on the primary outcome were missing at all time points, or if the tumour was classified as pure in situ. Patients planned for neoadjuvant systemic therapy were not present in the analytical cohort. The final study population comprised 2,906 patients.

Data collection

Data were obtained from the NKBC, which contains information on patient and tumour characteristics, diagnostics, surgical and oncological treatment, and longitudinal follow-up. PROMs have been prospectively collected within the register since 2020 via the INCA platform, using validated instruments including the EORTC QLQ-C30 and the EORTC

QLQ-BR23, at baseline prior to treatment and at 6 and 12 months after diagnosis. PROM data were available at baseline for all 2,906 included patients, at 6 months for 2,638 patients, and at 12 months for 2,678 patients.

The primary exposure was lead time from histopathological diagnosis to primary surgery, defined as the number of calendar days from confirmed diagnosis to primary breast surgery as recorded in the NKBC. The primary outcome was the response to EORTC QLQ-BR23 item 43, assessing worry about future health on a four-point ordinal scale. Secondary outcomes were general worry, tension, depressed mood, and overall quality of life, assessed using items from the EORTC QLQ-C30.

These outcomes were analysed using the original ordinal response categories of the selected EORTC items rather than transformed multi-item scale scores. Interpretation followed the direction of the original response categories for each item. The analyses examined associations within the study cohort according to diagnosis-to-surgery lead time and did not involve comparison with an external reference population.

Analysis

Lead time was standardised to a z-score so that estimated effects corresponded to a one standard deviation increase of approximately 19 days. Associations between lead time and each ordinal PROM outcome were estimated using mixed-effects cumulative link models, with random intercepts for patient and healthcare region. Because the study period coincided with the COVID-19 pandemic, during which both surgical lead times and population-level health-related worry may have differed from later years, the primary model included a binary pandemic-era indicator distinguishing 2020–2021 from 2022–2024. The model was additionally adjusted for age at diagnosis, PROM time point, clinical tumour size, clinical nodal status, histological grade, surgery type, and biomarker status. Three pre-specified sensitivity analyses were conducted: restriction to lead times of 90 days or fewer, restriction to T1 tumours, and a binary 35-day threshold contrast. Linearity of the exposure–outcome relationship was assessed using a categorical model and a generalised additive model with a natural cubic spline. Analyses were conducted using complete-case data in R version 4.5.1.
Baseline characteristics of patients contributing all three PROM time points were similar to those contributing fewer observations, suggesting limited within-cohort informative dropout. However, as baseline PROM coverage increased gradually across the study period, from approximately

5% of eligible cases in 2020 to 13% in 2024, units with more systematic PROM implementation may serve patient populations that differ systematically, and this source of selection cannot be fully excluded.

Results

The principal empirical findings across the four studies are summarised in Table 2 and described in greater detail below.

Table 2. Key empirical findings from Studies I–IV

Study	Primary result	Supporting result	Empirical interpretation
I	Adjusted TCT difference: -30.7 min	Differences across perioperative phases	CT had shorter perioperative process time
II	Mean cost saving: SEK 3,547 per operation	Bootstrap probability cost-saving 96.2%; annual saving approximately SEK 681,000	CT model was less costly within the perioperative episode
III	Significant SAQ-OR differences in safety climate, stress recognition and management perception	Four qualitative categories identified	Organisational conditions differed in ways experienced by staff
IV	Lead time associated with future-health worry, OR 1.15 per SD	No association with general worry, tension, depressed mood, QoL	Waiting time related specifically to future-health worry

Note. OR, odds ratio.

Study I

Among the included 526 patients undergoing elective surgery for breast cancer or malignant melanoma, total case time was significantly shorter at the cardiothoracic surgery department than at the general surgery department. The unadjusted mean difference was 40.3 minutes (206.9 vs

166.6 minutes, general surgery vs cardiothoracic surgery). Differences were observed across all three phases of the perioperative process: room set-up and preparation (71.3 vs 52.1 minutes), procedure time (97.8 vs 84.6 minutes), and post-procedure clean-up (38.1 vs 29.9 minutes).

The two departments differed in several organisational features, including anaesthesia staffing, nursing skill mix, team working hours, and access to auxiliary personnel outside the operating room. To distinguish these organisational features from case-level differences, available patient-, procedure-, and surgeon-related variables were reviewed according to whether they were determined before or after assignment to the operating department. A linear regression model was then constructed using pre-assignment variables selected on the basis of clinical relevance and plausible confounding, thereby accounting for measured differences in case mix between the departments where data were available in full (Table 3).

Table 3. Variables and operational definitions used in Study I

Domain	**Variable**	**Operationalisation in Study I**	**Relevance to thesis aim**
Patient characteristics	Age, BMI, ASA score, sex	Extracted from Orbit/electronic records	Adjustment / contextual comparison
Case and procedural characteristics	Procedure group	Categorised as extensive breast cancer, limited breast cancer, or malignant melanoma	Adjustment for case mix
Organisational setting	Surgical department	General surgery department or cardiothoracic surgery department	Main comparison of organisational context

Domain	Variable	Operationalisation in Study I	Relevance to thesis aim
Surgeon-related case characteristics	High-volume surgeon; consultant or surgical trainee present	High-volume surgeon defined as >50 relevant procedures per year; consultant status assessed for the main surgeon	Adjustment for surgeon-related factors
Perioperative workflow outcome	Total case time	Room set-up start until room clean-up finish	Primary productivity outcome

Note. Extensive breast cancer was defined as C50.* with any of HAC22, HAB** + PJD42, or PJD42. Limited breast cancer included C50.* with other combinations than those included in the extensive group. Malignant melanoma was defined according to the primary ICD-10 classification. Certain interventions, such as PJA10 and ZZR70, could occur in any procedure group. High-volume surgeon was defined as performing >50 relevant procedures per year. Consultant status was assessed only for the main surgeon.

After adjustment for patient characteristics, procedure group, and surgeon-related factors in a linear regression model, total case time remained significantly shorter in the cardiothoracic department than in the general surgery department, with an estimated difference of −30.7 minutes (95% CI −40.0 to −21.3; $p<0.001$). This effect was consistent across all three procedure groups and robust to alternative model specifications and imputation strategies for missing BMI values. The two departmental sub-cohorts represented real-world organisational settings rather than randomly assigned comparison groups. Patient demographics were broadly similar between departments, and the same group of surgeons operated at both sites, while procedure mix and some case characteristics differed descriptively. These descriptive differences were used as contextual information for interpreting the adjusted analysis and should not be understood as formally tested group differences.

Student-related variables were also explored, as student presence was considered a clinically plausible explanation for longer operating room times. These variables included the number and type of students present

in the operating room. Including student-related variables did not meaningfully change the estimated department effect, and they were therefore not retained in the final model. This decision was further supported by uncertainty regarding whether student allocation occurred before or after assignment to operating department. Because malignant melanoma cases were overrepresented in the general surgery department, additional sensitivity analyses were performed using interaction terms between department and procedure group. No statistically significant interaction effects were observed, and inclusion of these terms did not substantially alter the main results.

Study II

Study II extended the findings from Study I by examining whether the observed difference in perioperative process time was associated with differences in cost, capacity, and waiting time. In this bottom-up micro-costing and cost-minimisation analysis of 543 consecutive patients undergoing elective breast cancer surgery, the cardiothoracic surgery department was less costly than the general surgery department in 96.2% of bootstrap samples. The mean cost saving was SEK 3,547 per operation (95% CI −674 to 7,510).

The cost difference was not explained by lower staffing intensity at the cardiothoracic department. On the contrary, personnel costs were higher in the cardiothoracic setting due to constant anaesthesiologist presence. However, these higher personnel costs were offset by lower costs in other components, particularly medication costs, which were higher in the general surgery department and accounted for the largest share of the total cost difference. Thus, Study II suggested that the organisational model influenced cost through the combined effect of staffing configuration, time use, and perioperative resource consumption rather than through staffing levels alone. This pattern is illustrated in Figure 6.

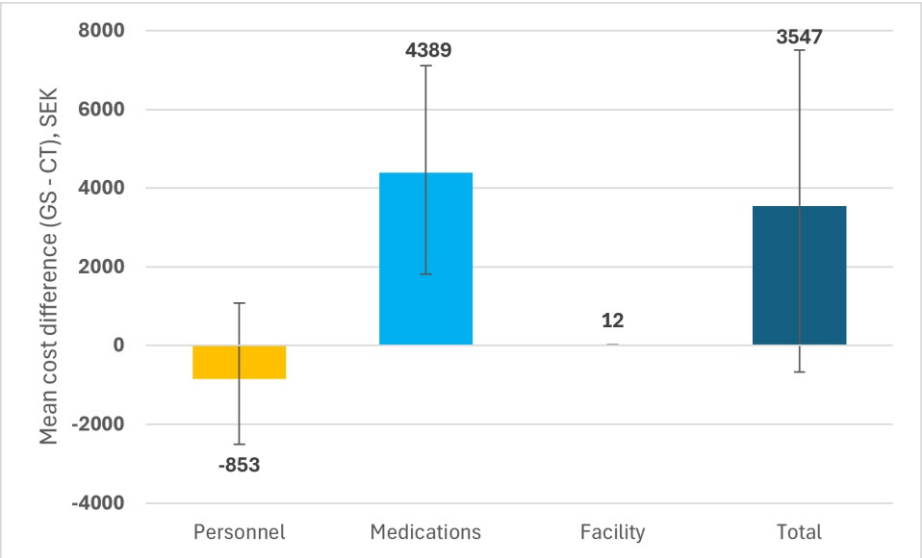


Figure 6. Decomposition of mean perioperative cost difference between departments in Study II. Values are presented as GS – CT, such that positive values indicate higher costs in the general surgery department and negative values indicate higher costs in the cardiothoracic surgery department. Error bars represent 95% bootstrap confidence intervals.

The capacity analysis further showed that the shorter case times observed at the cardiothoracic department translated into greater theoretical operating room productivity. Procedure-specific efficiency ratios, calculated as general surgery total case time divided by cardiothoracic total case time, ranged from 1.26 to 1.30 across procedure types. Based on a nine-hour operating day, this corresponded to 3.2 operations per room-day at the cardiothoracic department compared with 2.4 at the general surgery department. To manage the observed patient inflow, the corresponding break-even capacity was 1.17 room-days per week for the cardiothoracic model and 1.52 room-days per week for the general surgery model.

Finally, the deterministic simulation suggested that the cardiothoracic department contributed to maintaining shorter preoperative waiting times. When cardiothoracic operating room capacity was removed and the general surgery department was assumed to absorb the full patient volume at unchanged efficiency, mean preoperative waiting time increased by 15.6%, from 39.0 to 45.1 days. At the observed annual volume, the projected annual saving was approximately SEK 681,000, increasing to approximately SEK 1.77 million at 500 patients per year. The

productivity difference identified in Study I therefore also had measurable implications for cost, capacity, and simulated waiting time.

Study III

Study III used a sequential explanatory mixed-methods design to examine how operating room nurses and anaesthesia nurses at the same two departments perceived productivity and what organisational conditions supported or thwarted a sustainable pace over time. In the initial SAQ-OR survey of 47 staff members, significant between-unit differences were observed in three of six domains: safety climate (Unit A 32% vs Unit B 68%; p=0.019), stress recognition (57% vs 21%; p=0.018), and perceptions of management (18% vs 63%; p=0.002). Teamwork climate, job satisfaction, and working conditions did not differ significantly between units. The SAQ-OR results should therefore be interpreted as contextual signals of perceived safety climate and organisational conditions, used to inform the focus and purposive sampling of the subsequent interview phase, rather than as direct measures of productivity or definitive rankings of the two departments.

Qualitative content analysis of 24 interviews identified four cross-cutting categories describing conditions that supported or undermined sustainable operating room productivity. The first, predictable workflow and usable tools, captured how reliable equipment availability, coherent information systems, and stable staffing were associated with steadier tempo, fewer workarounds, and earlier escalation. The second, psychologically safe teamwork, described how civil tone, mutual help, and equitable voice across professional roles supported timely correction and collaboration, whereas hierarchy, silence, and sharp exchanges were associated with delayed correction and reduced safety. The third, participatory planning and visible follow-through, reflected the importance of genuine staff involvement in list planning, clear rationales for changes, and visible managerial response to raised concerns. The fourth, learning structures for mastery, described how supervision, reflection, and protected learning supported competence development, while sustained workload without recovery contributed to motivational fatigue and thoughts of leaving.

Using Self-Determination Theory as an abductive interpretive framework, these categories were interpreted as organisational conditions that supported or thwarted the basic psychological needs for autonomy, competence, and relatedness (Figure 7). Across the categories, descriptions of supportive organisational conditions contrasted with

accounts of fragmented information, limited influence over planning, interpersonal strain, and compensatory workarounds. These findings show that staff experiences of sustainable pace depended on organisational conditions as well as on time and resources.

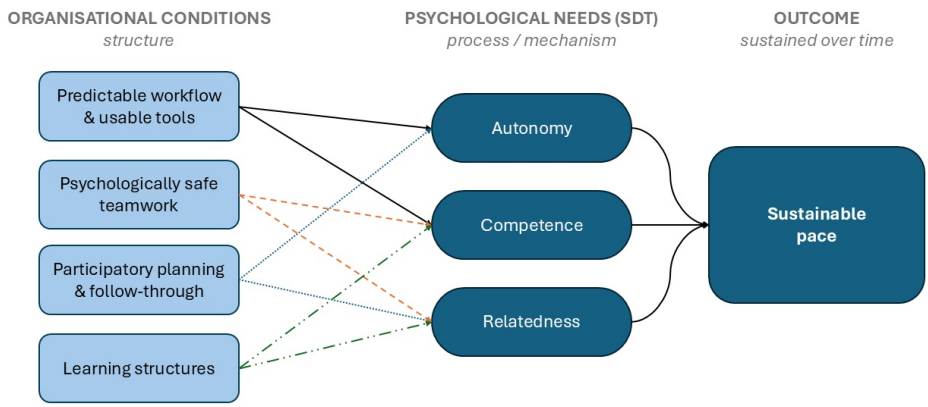


Figure 7. Conceptual model derived from the Study III findings. Organisational conditions are interpreted as supporting satisfaction of the three basic psychological needs described in Self-Determination Theory: autonomy, competence, and relatedness. Together, these need-supportive conditions underpin a sustainable working pace. Line style and colour identify the originating organisational condition. Arrows denote supportive relationships grounded in the qualitative analysis and should not be interpreted as statistically tested causal paths.

Study IV

Study IV examined whether diagnosis-to-surgery lead time was associated with patient-reported psychological outcomes in women with primary invasive breast cancer. In this longitudinal Swedish registry-based cohort study, 2,906 women undergoing primary surgery between January 2020 and December 2024 were included, with PROM observations at baseline (n=2,906), 6 months (n=2,638), and 12 months (n=2,678). Median age at diagnosis was 63 years (IQR 54–71), and median diagnosis-to-surgery lead time was 32 days (IQR 23–42). Most tumours were classified as T1 (70.1%), and 83.0% of patients underwent breast-conserving surgery. Overall, 61.6% were operated within 35 days. Substantial regional variation was observed, with the proportion operated within 35 days ranging from 77.6% in Region North to 56.8% in Region South. A temporal trend was also evident: median lead time increased from 25 days in 2020 to 34.5 days in 2024, while the proportion operated within 35 days declined from 78.1% to 54.8%.

Longer lead time was associated with greater worry about future health. In the primary mixed-effects cumulative link model, each one-SD increase in lead time, corresponding to approximately 19 days, was associated with higher odds of reporting greater worry about future health (adjusted odds ratio (OR) 1.15, 95% CI 1.04–1.27; p=0.008). At baseline, 42.2% of patients with lead times exceeding 35 days reported substantial worry about future health, compared with 35.1% among those operated within 35 days. This 35-day comparison was interpreted descriptively, as the pre-specified binary threshold model did not converge. The adjusted associations between diagnosis-to-surgery lead time and the five patient-reported outcomes are summarised in Figure 8.

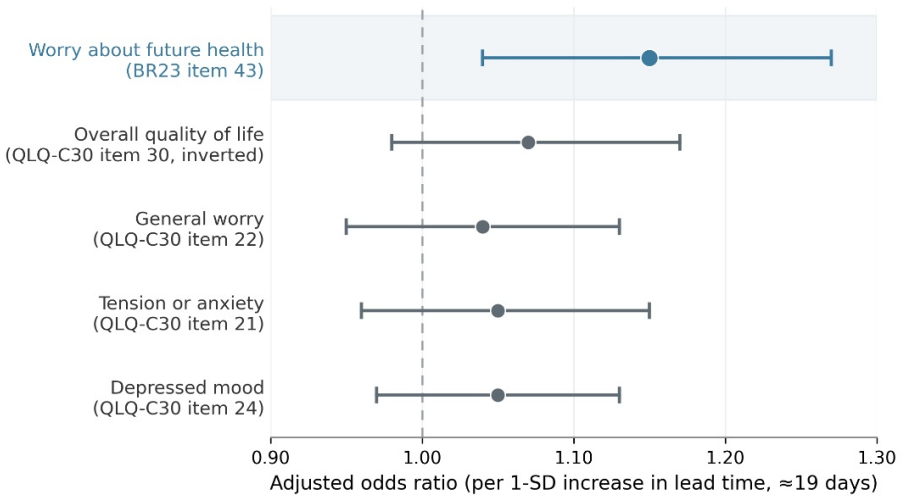


Figure 8. Summary of adjusted associations between diagnosis-to-surgery lead time and patient-reported outcomes in Study IV. Odds ratios are shown per one standard deviation increase in lead time, corresponding to approximately 19 days. Values above 1 indicate higher odds of a worse patient-reported outcome. Error bars represent 95% confidence intervals.

No corresponding statistically significant associations were observed for general worry, tension or anxiety, depressed mood, or overall quality of life. This pattern suggested that lead time was associated specifically with future-oriented health worry rather than with a general increase in psychological distress. The association remained significant when the analysis was restricted to lead times of 90 days or fewer (OR 1.14, 95% CI 1.04–1.26; p=0.007), and categorical and spline-based analyses supported an approximately linear rather than threshold-based relationship across the lead-time distribution.

Discussion

General discussion of the main findings

Previous research has suggested that operative practices, such as personnel management, case scheduling, and performance measurements, may have a greater influence on surgical productivity than broader strategic characteristics such as unit size or academic status, and that different strategic positions may require different operative practices (85). Our findings partly support this view, but also suggest that the relationship is more complex. In Study I, total case time was substantially shorter in one organisational setting than in the other, and Study II showed that this difference was also reflected in costs and capacity. Table 4 summarises the principal findings, their organisational interpretation, and their contribution to the thesis-level synthesis across the four studies overall.

Table 4. Main findings, organisational interpretation, and thesis contribution of Studies I–IV

Study	Main finding	Organisational interpretation	Contribution to the thesis
I	TCT was shorter in the cardiothoracic department	Departmental organisation was associated with perioperative process time	Links organisation to perioperative productivity
II	The more efficient model was less costly and eased capacity and waiting time pressures	Process efficiency translated into resource use and capacity	Extends productivity to system-level outcomes

Study	Main finding	Organisational interpretation	Contribution to the thesis
III	Sustainable pace depended on predictable workflow, psychological safety, participatory planning, and learning structures	Staff experience reflected the organisational conditions underlying dependable performance	Adds the workforce perspective
IV	Longer lead time was associated with greater worry about future health	Waiting time was meaningful from the patient perspective	Links timely access to patient-reported outcomes

Note. The table summarises the principal thesis-level interpretation of each study rather than all study-specific results. PROM, patient-reported outcome measure.

Organisation as a condition for perioperative productivity and resource use

Studies I and II indicate that perioperative productivity is related to organisational conditions extending beyond formal departmental design. The organisational differences between the two departments in Studies I and II were primarily structural: staffing ratios, anaesthesiologist presence, skill mix, and team continuity differed systematically between settings. These differences were associated with shorter total case time, lower perioperative costs, and greater operative capacity. The observed productivity gap was therefore more consistent with differences in organisational configuration than with the measured differences in individual performance or case complexity. This interpretation is also consistent with the systematic review by Pasquer et al., which identified team composition, team stability, teamwork, work scheduling, and disturbing elements in the operating room as factors that may influence surgical performance (38). These findings support a broader view of productivity in which organisational structure and operative practices interact. Similar concerns have been raised regarding organisational silos in perioperative care, where separately managed departments and conflicting local incentives may fragment the surgical pathway and limit the effects of isolated process improvements (86). In a high-throughput

joint arthroplasty setting, deliberate redesign of the perioperative process, including parallel rather than strictly sequential activities, produced sustained gains in operating room throughput (87).

Workforce stability is a critical but often underappreciated determinant of sustainable OR throughput. Stable, experienced teams may develop familiarity and refined communication patterns that contribute to more predictable procedures. This interpretation is also supported by a systematic review of dedicated surgical teams, which reported favourable findings for turnover time, teamwork, communication, and costs, although the underlying evidence was heterogeneous (88). Beyond speed, stable teams preserve institutional knowledge regarding protocols, equipment, and surgeon preferences, expertise that is both difficult to quantify and costly to replace through recruitment and retraining.

Workforce instability also has implications for system resilience. Aggressive efficiency measures that minimise idle time and reduce staffing buffers may increase clinician stress and reduce job satisfaction, contributing to burnout and attrition, which in turn further erodes capacity (23). This dynamic illustrates a broader tension in perioperative management: optimising for short-term utilisation metrics may undermine the organisational conditions necessary for sustained performance. Team stability should therefore be understood not only as a staffing issue, but as one organisational condition that may link efficiency, workflow reliability, and care quality (38).

However, organisational efficiency cannot be fully understood through time and cost alone, as the same structural conditions also shape how work is experienced and sustained by staff.

Staff experience and the conditions for sustainable pace

Study III showed that sustainable pace depended partly on predictable workflow and access to usable tools and coherent information. Participants described how fragmented documentation, unavailable equipment, and unclear information flows interrupted work and generated avoidable workarounds. These findings are consistent with research on team situation awareness, showing that open, non-directed, or delayed communication loops may increase susceptibility to information loss in the operating room (89).

Participants in Study III also described more opportunities for social interaction at one department, both during and outside working hours.

Such practices may have contributed to greater interprofessional familiarity and a more cohesive departmental climate. Although this was only one aspect of the qualitative accounts, it is consistent with literature suggesting that team familiarity can support OR efficiency. Witmer et al. reported in a systematic review that operative team familiarity was associated with shorter operating time (90), while Stucky and De Jong's integrative review linked surgical team familiarity to fewer errors, disruptions, miscommunication, and readmissions, as well as improved team member safety (91). Team familiarity should therefore be understood not merely as a social feature of workplace culture, but as a potential organisational resource for surgical performance. Swedish qualitative research has similarly suggested that differing professional rationalities and insufficient social and organisational support may create communication thresholds, while organisational arrangements that facilitate cross-professional interaction can support shared understanding and trust (92).

During the first phase of Study III, the between-unit differences in the survey were interpreted as contextual signals rather than as an outright ranking of the two departments. The observed differences in safety climate, stress recognition, and perceptions of management could be partly understood through the categories identified in the qualitative phase: where predictable workflow, usable tools, psychologically safe teamwork, participatory planning, and structures for learning and mastery were more consistently narrated, the survey results also suggested more favourable signals in safety climate and management perceptions, alongside the observed contrast in stress recognition. Conversely, where need-thwarting patterns were more prominent, including late list changes, tokenistic consultation, parallel documentation, and sharp tone, the qualitative accounts more often reflected workaround culture, reduced initiative, and conditional retention. Self-Determination Theory offered a particularly apt frame for this context, given that OR work is characterised by high interdependence, time pressure, and the simultaneous need for technical precision and interprofessional coordination, conditions that make support for autonomy, competence, and relatedness particularly relevant (15, 78). As illustrated in Figure 7, the findings of Study III suggest that sustainable pace is not produced by individual motivation alone, but by organisational conditions that support autonomy, competence, and relatedness. Staff need meaningful influence over workflow decisions and visible responses when concerns are raised. Predictable work and opportunities to learn also mattered, as did relationships in which staff could speak up when problems arose.

Study III offers one possible organisational interpretation of the differences observed in Studies I and II: everyday conditions for coordination and staff motivation may contribute alongside formal staffing and scheduling structures. Where such conditions are stronger, teams may be better able to coordinate fluidly, anticipate needs, and maintain pace without relying primarily on informal compensatory strategies. The workaround culture described in Study III may instead indicate that staff were compensating for organisational pressure. When workarounds become central to maintaining flow, they may indicate that the slack necessary for safe and sustainable performance has been reduced, and that the system is more fragile than output metrics alone suggest. This interpretation is consistent with systematic review evidence showing that operating room disruptions and interruptions are associated with longer operative time and poorer team performance, as well as with some patient-safety outcomes (15).

The categories identified in Study III resonate with Edmondson's foundational work on psychological safety, which showed that teams in which members feel safe to speak up, ask questions, and raise concerns without fear of interpersonal consequences are more likely to engage in learning behaviour (93). In the OR context, where silence may affect not only team functioning but also patient safety and workflow, psychological safety is therefore not merely a motivational concern, but an operational one.

The relationship between staff experience and sustainable perioperative productivity is also supported by related evidence from the broader workforce literature, where nurse burnout has been associated with nurse-reported quality of care and patient outcomes (94). In perioperative settings, work environment, staffing and resources, emotional exhaustion, workload, leadership, and collegial relationships have been linked to job dissatisfaction, intention to leave, and retention (95, 96). Staff experience should therefore not be treated as separate from productivity: when the conditions that sustain motivation and coordination are eroded, the consequences may not be visible in throughput metrics alone, but may emerge over time as retention problems, workaround dependency, and reduced organisational resilience.

This points to a broader distinction that the findings of this thesis collectively illustrate: the difference between short-term throughput and sustainable productivity. Flow can be optimised in a narrow sense by filling OR slots, reducing turnover time, and maximising case volume.

However, such gains may be achieved through intensified effort and informal compensation if the underlying organisational conditions are weak. Sustainable productivity, by contrast, requires conditions that support staff over time: appropriate distribution of workload, sufficient operational slack, protected learning structures, and interprofessional relationships that enable coordination under pressure.

The retention signals identified in Study III warrant particular attention in this context. Conditional commitment also has implications for organisational capacity. Motivational fatigue and intention to leave among perioperative nurses reduce the stability of the workforce on which OR function depends. As noted above, international perioperative workforce data suggest that vacant full-time perioperative nursing positions increased from 3.1% to 18% between 2013 and 2023, and that the proportion of managers reporting moderate- to crisis-level shortages ranged from 37% to 73% (34). Although these figures are not Sweden-specific, they illustrate a broader workforce instability that is directly relevant to surgical capacity. Study III adds to this picture by illustrating plausible local mechanisms through which retention pressure may be produced and sustained in everyday work: how planning is conducted, how concerns are handled, and how learning is supported. Workforce instability therefore extends beyond a human resources concern and may directly threaten the capacity and resilience of the perioperative system.

The organisational conditions examined across Studies I to III shape efficiency, staff experience, and the capacity to provide timely surgical care. A system that absorbs variation through informal staff effort rather than organisational design may maintain output in the short term, but risks reducing the adaptive capacity needed to prevent delays, cancellations, and unpredictability in patient access (97). For patients awaiting breast cancer surgery, such delays are not merely logistical inconveniences. They occur during a period already characterised by fear, uncertainty, and concern about future health. Study IV therefore extends the analytical lens from the organisation and its staff to the patient, examining whether the time between diagnosis and surgery carries measurable consequences for patient-reported outcomes.

Waiting time and patient-reported psychological outcomes

In Study IV, longer diagnosis-to-surgery lead time was associated with greater worry about future health, while no corresponding association was observed for general worry, tension, depressed mood, or overall quality of

life. This outcome-specific pattern suggests that waiting time may primarily affect disease-related cognitive appraisal rather than producing a general deterioration in emotional wellbeing, at least within the time frame and PROM domains examined. However, such worry should not be regarded as a purely subjective or isolated patient-reported outcome. In newly diagnosed patients with breast cancer, anxiety, depression, and stress reaction or adjustment disorders have been associated with higher healthcare resource utilisation, including more emergency department visits, inpatient admissions, primary care visits, specialist visits, and higher all-cause healthcare costs during the first year after diagnosis (98). More broadly, a systematic review found that anxiety disorders and health anxiety are associated with increased healthcare utilisation across multiple care settings (99).

These findings suggest that prolonged uncertainty before surgery may have system-level implications beyond patient experience alone. Although the present thesis cannot determine whether waiting-related worry leads to somatic morbidity, prior evidence suggests that acute psychological distress around major health events may have physical consequences. At the extreme end of this spectrum, intense emotional or physical stress is a recognised trigger of stress-associated cardiac conditions such as Takotsubo syndrome, although no such consequence can be inferred from the findings of Study IV (100). Following a cancer diagnosis, a large Swedish registry study found an immediate increase in cardiovascular death, a pattern consistent with the potential effects of acute psychological stress (101). Stress-related disorders have also been linked to subsequent cardiovascular disease in a population-based sibling-controlled cohort study (102). These studies should not be read as evidence that the associations observed in Study IV cause later morbidity, but they strengthen the argument that avoidable uncertainty before surgery is clinically and organisationally relevant. Reducing unnecessary waiting and providing clear preoperative communication may therefore have implications for patient-reported wellbeing, healthcare demand, and the resilience of surgical pathways.

Linking structure, process, and outcomes across levels

Taken together, the four studies can be interpreted as a multilevel chain linking organisational structure, perioperative processes, and outcomes for both staff and patients. In Studies I and II, structural differences between the two operating departments were reflected in measurable process and resource outcomes, including shorter total case times, lower perioperative costs, and potential effects on capacity and waiting time.

Study III added an explanatory layer by showing how everyday organisational processes, such as planning, communication, learning structures, psychological safety, and the handling of disruptions, shaped staff experience and the conditions for sustainable pace. Study IV then extended the outcome perspective to patients, showing that longer lead time before surgery was associated with greater worry about future health.

Productivity, staff sustainability, and patient-reported outcomes should therefore not be understood as separate domains, but as interconnected consequences of how perioperative care is organised.

This interpretation also highlights a potential vulnerability in highly pressured surgical systems. When organisational slack is progressively reduced, teams may initially preserve output through intensified effort, informal compensation, and local workarounds. However, maintained throughput under such conditions does not necessarily indicate that the system is functioning sustainably. It may instead reflect that variation is being absorbed by staff effort rather than by resilient organisational design. Over time, this can reduce the system's capacity to respond to disturbances without consequences for staff wellbeing, workflow reliability, or patient access.

Surgical organisation should therefore not be understood as a background condition, but as an organisational condition that shapes how care is produced and experienced. Structures such as staffing models, workflow design, and departmental organisation influence the processes through which teams coordinate work, absorb variation, and maintain flow. These processes, in turn, are linked to outcomes that extend beyond conventional productivity metrics. A productive surgical system is therefore not simply one that performs more procedures, but one that does so through structures and processes that remain safe, coordinated, and sustainable over time (32).

Methodological considerations

Study population

The four studies included different but complementary study populations, reflecting the multilevel aim of the thesis. Study I was based on patients undergoing breast cancer or malignant melanoma surgery, while Study II focused specifically on breast cancer surgery. Both studies were conducted within one Swedish university hospital, where the same surgeon group performed the relevant procedures in two differently organised operating

departments. This was an important methodological strength, as operative time is influenced not only by procedure type and patient factors, but also by surgeon-related variation. Previous research has identified the surgeon as an important contributor to variability in surgical procedure times, although procedure type often remains the dominant driver of operative duration (103). By comparing departments in which the same surgeon group performed the same types of procedures, Studies I and II reduced a key source of variation that often limits organisational comparisons in surgical care. This strengthened the internal comparison and increased the plausibility that observed differences in time and cost reflected departmental organisation rather than differences between surgeon groups. At the same time, the single-centre design and the specific organisational configuration limit the extent to which the observed time and cost differences can be generalised to other hospitals, surgical specialties, or operating departments with different staffing models, case-mix, or scheduling routines.

Study III included perioperative staff from the same hospital context, but at a later time point and with a different empirical focus. The staff population was therefore not used to explain the findings of Studies I and II causally, but to provide a mechanism-oriented understanding of how organisational conditions may be experienced and sustained in everyday OR work. The inclusion of operating room nurses and anaesthesia nurses enabled detailed exploration of frontline perspectives on workflow, coordination, and sustainable pace. However, the absence of other professional groups, including surgeons, assistant nurses, managers, and administrative planning staff, from the interview phase means that the findings primarily reflect nursing perspectives on perioperative organisation.

Study IV differed from the preceding studies by using national registry data from women with primary invasive breast cancer who underwent primary surgery and had available PROM data. This broadened the empirical scope beyond the single-centre setting and enabled analysis of patient-reported outcomes in a large national registry-based cohort. However, the cohort does not represent all patients with breast cancer, as patients with in situ disease, previous breast cancer, implausible lead times, defined in Study IV as intervals shorter than 1 day or longer than 365 days, and those planned for neoadjuvant systemic therapy were excluded. In addition, PROM participation may introduce selection mechanisms that limit the generalisability of patient-reported findings to all operated patients.

Taken together, the study populations support different types of inference: Studies I and II provide internally comparable organisational comparisons, Study III provides transferable insights into staff experience and organisational mechanisms, and Study IV provides population-level evidence on the patient perspective. The findings should therefore be interpreted as complementary rather than directly interchangeable across populations.

Data validation

The four studies relied on different types of data, and the validation challenges therefore differed across the thesis. The secondary use of routinely collected perioperative and registry data entails methodological and reporting considerations that differ from those of data collected specifically for research purposes (104). In Studies I and II, the main empirical basis consisted of routinely registered perioperative data from Orbit and related operating room records. Such data are valuable because they reflect everyday clinical practice and are generated as part of the ordinary perioperative workflow. At the same time, secondary use of routine clinical data is dependent on the quality of the original registrations. In this sense, the data could not be more precise than the timestamps entered into Orbit. This is a general limitation of routine clinical data, for which completeness, correctness, concordance, plausibility, and currency are commonly recognised dimensions of data quality (105). To reduce the influence of obvious registration problems, cases with clear time-registration errors or missing key values were excluded. However, the underlying timestamps themselves could not be independently verified against an external gold standard.

The large number of operations across several years may have reduced the influence of occasional random registration errors. More systematic registration practices, such as consistently using standardised intervals for specific perioperative phases, could still affect the absolute precision of individual time estimates. However, if such practices were applied consistently within the ordinary workflow, they may still capture meaningful operational boundaries and support comparison of overall patterns between departments. The main concern would be differential registration practices between departments, which cannot be fully excluded. For this reason, the time estimates should be interpreted as structured measures of registered perioperative processes rather than exact representations of every minute of clinical activity.

In Study II, data validation also concerned the accuracy of resource-use and cost estimates. The bottom-up micro-costing approach required combining routinely registered time data with information on staffing, salaries, medications, consumables, and facility costs. Direct observations and operating room records strengthened the mapping of resource use, while the consistency between the cost analysis and the time differences observed in Study I supported the overall plausibility of the organisational comparison. Nevertheless, several cost components required standardised assumptions, particularly where individual-level variation in resource use could not be fully observed. The cost estimates should therefore be interpreted as carefully constructed approximations of perioperative resource use rather than exact accounting values.

In Study III, data validation concerned both the survey and the qualitative material. The SAQ-OR was used not as a definitive measure of unit performance, but as a structured mapping tool to identify contextual signals that could guide the qualitative phase. The survey findings were therefore used to inform interview sampling and probing, while the interviews provided the mechanism-oriented interpretation. Because survey denominators were based on staffing records rather than confirmed individual receipt of the questionnaire, and because no systematic information on non-respondents was available, the SAQ-OR results were interpreted as approximate contextual signals rather than representative estimates of unit climate. The qualitative data in Study III were strengthened through verbatim transcription, systematic coding, reflexive discussion, and repeated movement between codes, categories, and the underlying data. The use of an inductive content analysis followed by an abductive Self-Determination Theory overlay made it possible to preserve the empirical content of the interviews while also interpreting patterns through a theoretical lens. However, the initial coding was primarily conducted by one researcher, and formal member checking was not performed. To support analytical consistency, one transcript was independently double-coded with a second researcher to calibrate the coding approach. These limitations were otherwise addressed through co-author discussions, category refinement, and transparency regarding the researcher's insider position.

Study IV relied on data from the Swedish National Quality Register for Breast Cancer, which provided national clinical data and repeated PROM observations. The exposure was defined using registry-recorded dates for histopathological diagnosis and primary surgery, and implausible lead times, defined as intervals shorter than 1 day or longer than 365 days, were excluded to reduce the influence of likely registration errors or atypical

clinical pathways. The PROM outcomes were based on established EORTC items, which strengthened measurement validity. However, PROM participation was not random, particularly at baseline, and implementation varied between reporting units during the study period. The inclusion of pandemic years, during which both waiting times and registry participation may have been affected by service disruptions, represents an additional source of variability that could not be fully controlled for. This means that the clinical registry data were strong, but the patient-reported cohort may still be affected by selection mechanisms. Sensitivity analyses, including restrictions on lead time and alternative calendar-time adjustment, were therefore important for assessing the robustness of the findings.

Taken together, the data sources were appropriate for the different aims of the four studies, but each required study-specific validation strategies. The quantitative studies benefited from routine clinical and registry data, while the qualitative study relied on transparency, reflexivity, and systematic analytical procedures. The main methodological implication is that the findings should be interpreted in relation to the type of data from which they were derived: time and cost estimates as structured measures of perioperative processes, qualitative accounts as mechanism-oriented interpretations of staff experience, and PROM data as patient-reported outcomes shaped by both clinical pathways and response patterns.

Transferability and generalisability

The exact magnitude of the observed time and cost differences should be interpreted in relation to the local organisational context. However, the design of Studies I and II, in which the same surgeon group operated across both departments, strengthens the transferability of the underlying organisational question: how departmental staffing, workflow, and coordination structures shape perioperative performance when surgeon-related variation is partly held constant. The findings should therefore not be read as evidence that the same time or cost differences would necessarily occur in other hospitals or specialties, but rather as evidence that organisational configuration can matter even when surgical expertise and clinical pathway are relatively similar.

For Study III, transferability lies less in the specific contrast between the two units and more in the mechanisms identified across staff accounts. Predictable workflow, usable tools, genuine participation in day-to-day planning decisions, psychological safety, and learning structures are not unique to the included hospital but represent organisational conditions

that are likely to be relevant in other perioperative settings. The qualitative findings are therefore best understood as transferable hypothesis-generating design principles rather than as a direct comparison of unit performance or as evidence of generalisable unit effects.

Study IV, by contrast, used national registry data and therefore has broader population-level relevance for Swedish breast cancer care. However, generalisability is still shaped by the inclusion criteria, PROM participation, and the exclusion of patients planned for neoadjuvant systemic therapy. The findings are therefore most applicable to women with primary invasive breast cancer undergoing primary surgery with available PROM data, rather than to all patients with breast cancer. Additionally, response bias cannot be excluded, as patients who chose to participate in PROM data collection may differ systematically from those who did not, both in terms of clinical characteristics and psychological disposition.

Limitations

Several limitations should be considered when interpreting the findings. Studies I and II were observational and conducted within a single hospital context. Although the same surgeon group operated across both departments, reducing one important source of variation, patients were not randomly allocated to organisational models, but rather according to a 'first come, first served' principle. Residual differences in scheduling, case allocation, registration practices, and local workflow may therefore have influenced the observed time and cost differences.

The use of routinely registered perioperative data also means that the analyses were dependent on the quality of the original registrations. Cases with obvious time-registration errors were excluded, but the underlying timestamps could not be independently verified against an external gold standard. In Study II, several cost components required standardised assumptions, meaning that the estimates should be interpreted as carefully constructed approximations rather than exact accounting values.

In Study III, the extended interview period introduced the possibility of organisational change over time. This was considered in the interpretation of findings, which were treated as plausibility-oriented patterns rather than fixed unit characteristics. Because the two interview rounds coincided with the two units (Unit B in 2022, Unit A in 2025), unit, calendar time, and interview round are confounded in the qualitative

phase. The analysis was therefore deliberately oriented toward cross-cutting organisational mechanisms rather than toward comparing the two units, and the categories should be read as transferable conditions rather than as unit-specific effects. The interview sample also primarily reflected nursing perspectives, as surgeons, managers, assistant nurses, and administrative planning staff were not included in the qualitative phase. This limits the extent to which the findings capture the full range of professional perspectives on perioperative organisation. This limitation is relevant because perceptions of teamwork, communication, and situation awareness may differ substantially between professional groups within the operating room (106).

Study IV was limited by the observational registry-based design and by the availability of PROM data. Although models adjusted for several clinical and demographic factors, residual confounding cannot be excluded. Factors such as tumour biology, diagnostic complexity, socioeconomic circumstances, comorbidity, local pathway organisation, and patient preferences may influence both lead time and patient-reported outcomes, but could not be fully captured in the available data. PROM participation was not random, particularly at baseline, and variation in implementation between reporting units may have introduced selection mechanisms. The findings therefore support an association between lead time and worry about future health, but cannot establish that waiting time caused the observed patient-reported differences.

It should be emphasised that the four studies do not empirically trace a single causal pathway from a specific organisational model to patient-reported outcomes. Rather, the thesis uses the Donabedian framework as an organising heuristic to integrate findings from different levels of the surgical system. Studies I–III examine organisational structure, process, resource use, and staff experience within the same local perioperative context, whereas Study IV extends the analytical perspective to timely access to surgery and patient-reported worry within the national breast cancer pathway. The link between these levels should therefore be understood as conceptual and analytical rather than as a directly observed causal chain.

Future perspectives

The findings of this thesis suggest several directions for future research. First, future studies should examine whether the organisational patterns identified here are reflected in longitudinal indicators of workforce

stability and system strain. One question is whether recurring workarounds can function as early indicators of such strain, for example by preceding higher sickness absence, turnover, vacancies, reduced retention, missed breaks, overtime, cancellations, or increased reliance on temporary compensatory solutions. This would make it possible to study how perioperative systems approach capacity thresholds, particularly when maintained output depends increasingly on workarounds, staff overextension, and the erosion of organisational slack.

Second, future research could extend the analysis of perioperative teamwork to examine how relational conditions across the full interprofessional team shape both staff experience and operative flow. While teamwork in the OR has been studied extensively in relation to communication, safety, and non-technical skills, and the surgeon–anaesthesiologist dyad has received particular attention in relation to postoperative outcomes, with greater dyad familiarity being associated with lower postoperative morbidity in complex gastrointestinal cancer surgery (107), less attention has been paid to how familiarity and relational conditions across the wider team, including operating room nurses, anaesthesia nurses, assistant nurses, coordinators, and managers, influence sustainable pace and staff wellbeing. Previous studies have shown that perceptions of teamwork, communication, and collaboration differ between professional groups, and that communication, mutual respect, and relationship dynamics shape how surgical teams function in practice (108, 109). Studies combining interviews, observations, workload measures, and team familiarity metrics could therefore explore whether these relational conditions influence both staff wellbeing and operative flow. Instruments such as the Surgery Task Load Index (SURG-TLX) (110) or other task-load measures could be useful if combined with qualitative data, as perceived workload alone is unlikely to capture the organisational mechanisms through which work becomes sustainable or unsustainable.

Third, and most directly, the associations described in this thesis are observational and cannot establish whether deliberately altering organisational structure would improve perioperative performance or patient access. A natural next step is therefore to move from observing the structure–process–outcome chain to testing it, by evaluating whether redesigning staffing models, skill mix, team continuity, and scheduling routines produces measurable changes in process and outcome rather than merely co-varying with them. A stepped-wedge cluster-randomised design would be well suited to such an evaluation, as the intervention would operate at the level of departments rather than individual patients, allow each unit to serve as its own control as it crosses sequentially from

usual organisation to the redesigned model, and accommodate the logistical and ethical constraints of altering how an entire surgical service is staffed and run (111). To remain faithful to the multilevel logic of this thesis, such a trial should span the same analytical levels examined here, combining process and capacity indicators such as total case time, utilisation, cancellations, and waiting time with staff-reported measures of sustainable pace and, where the clinical context allows, patient-reported outcomes. This would shift the research question from whether organisation is associated with perioperative performance to whether changing organisation can improve the experience of both those who deliver and those who receive surgical care.

Clinical implications

The findings of this thesis have implications for clinical management and organisational development. Improvement efforts should consider scheduling not only as a technical optimisation problem, but as an integrated organisational process involving staff, rooms, equipment, downstream capacity, and patient access. Operations research literature similarly emphasises that operating room scheduling requires coordinated allocation of multiple resources, including surgical teams, nurses, anaesthesia teams, equipment, and recovery capacity, and that inappropriate staff timetabling may contribute to overtime, inefficiency, and repeated schedule changes when staff interests are not considered (112, 113). From a clinical perspective, this means that productivity initiatives should include staff-facing indicators alongside conventional output measures. Monitoring case volume, turnover time, and utilisation is not sufficient if these gains are achieved through reduced slack, informal compensation, or declining workforce stability.

Finally, Study IV highlights that organisational performance should also be evaluated from the patient perspective. Timely access to surgery is not only a logistical endpoint, but part of the patient experience of cancer care. Clinical improvement work should therefore consider whether changes in perioperative organisation are accompanied by changes in waiting time, preoperative communication, and patient-reported worry. Reducing avoidable waiting and uncertainty may be a clinically meaningful outcome in itself, while also supporting more resilient surgical pathways.

Conclusion

Overall, the findings indicate that perioperative productivity is shaped by organisational conditions extending beyond operating room throughput. Across the four studies, differences in surgical department organisation were associated with variation in total case time, perioperative costs, staff experiences of sustainable pace, and patient-reported worry during the waiting period before surgery. These findings suggest that surgical organisation is relevant to how efficiently care is delivered, how work is experienced by staff, and how patients experience timely access to care.

Study I demonstrated that, when the same surgeon group performed the same types of procedures in two differently organised departments, total case times were shorter in the cardiothoracic department than in the general surgery department. This suggests that organisational configuration may matter even when surgical expertise and clinical pathways are held relatively constant. Study II extended this finding through a bottom-up micro-costing analysis, indicating that the organisational model with shorter case times was also less costly per operation and had capacity implications for preoperative waiting time. Study III examined staff experiences of the organisational conditions underlying sustainable pace, identifying predictable workflow, usable tools, participatory planning, psychological safety, and learning structures as important conditions for maintaining dependable work. The absence of such conditions was reflected in workaround culture, reduced initiative, and conditional retention. Study IV extended the analytical lens to patients, finding that longer lead time before surgery was associated with greater worry about future health, suggesting that timely access to surgery is relevant to the patient experience of cancer care.

Efficient surgical care therefore requires structures that reduce unnecessary variation and resource waste, but also work environments that support staff autonomy, competence, and relatedness, allowing teams to maintain a dependable pace without eroding safety, learning, or retention. These findings are relevant beyond the specific hospital context in which they were generated: the organisational mechanisms identified here, including how staffing models, workflow design, and interprofessional conditions shape both efficiency and sustainability, may represent transferable principles for perioperative improvement. From this perspective, productivity, workforce sustainability, and patient experience are not competing objectives, but interdependent dimensions of high-quality perioperative care.

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